DISC-NET ML Workshop

What Just Happened?

Architectures, Generative AI, LLMs

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- But how did it happen?

AlexNet

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- AlexNet was a reimplementation of CNNs first invented in the the late 1990s by Yann LeCun
- What distinguishes AlexNet was it was implemented on two GPUs and trained on 1.3 million large images
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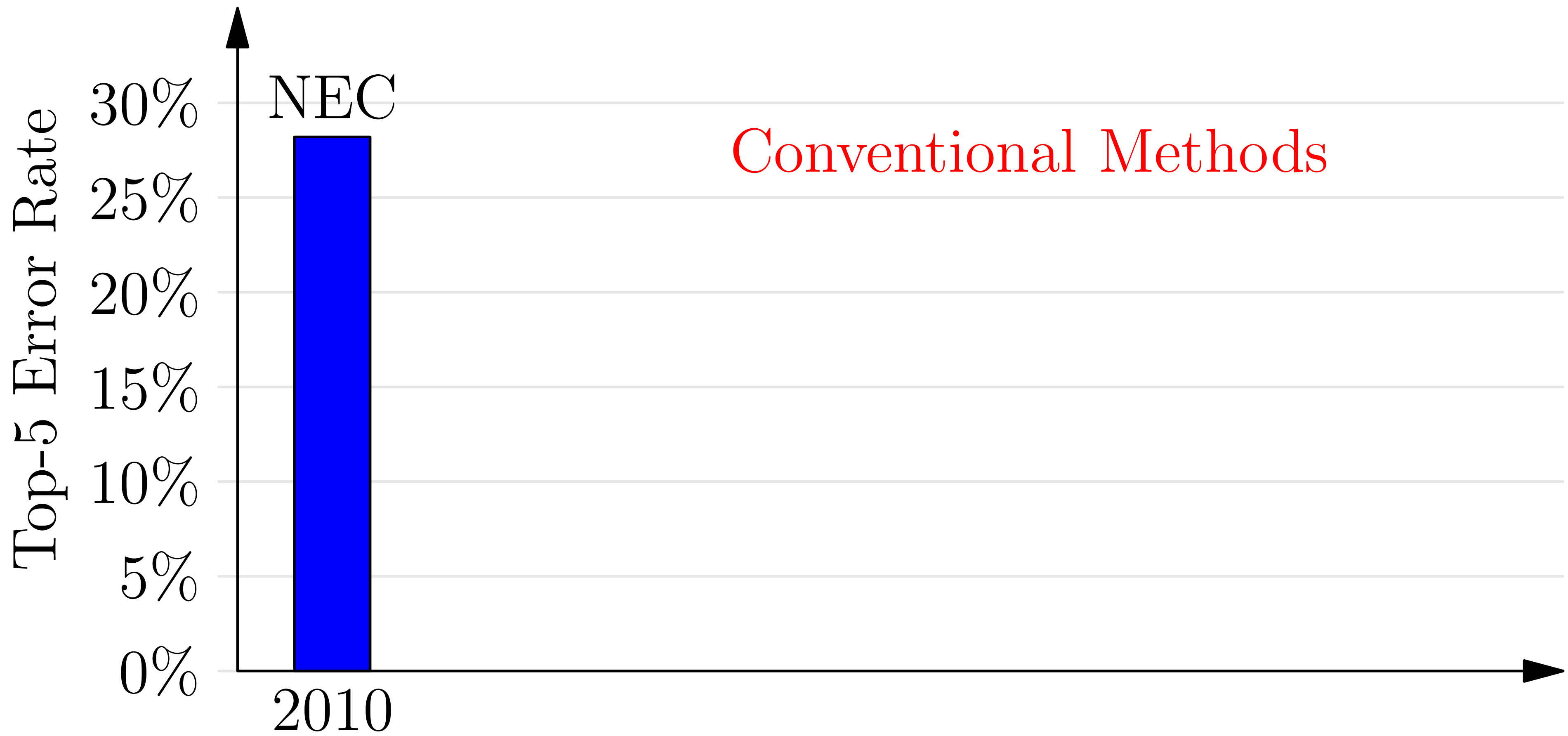

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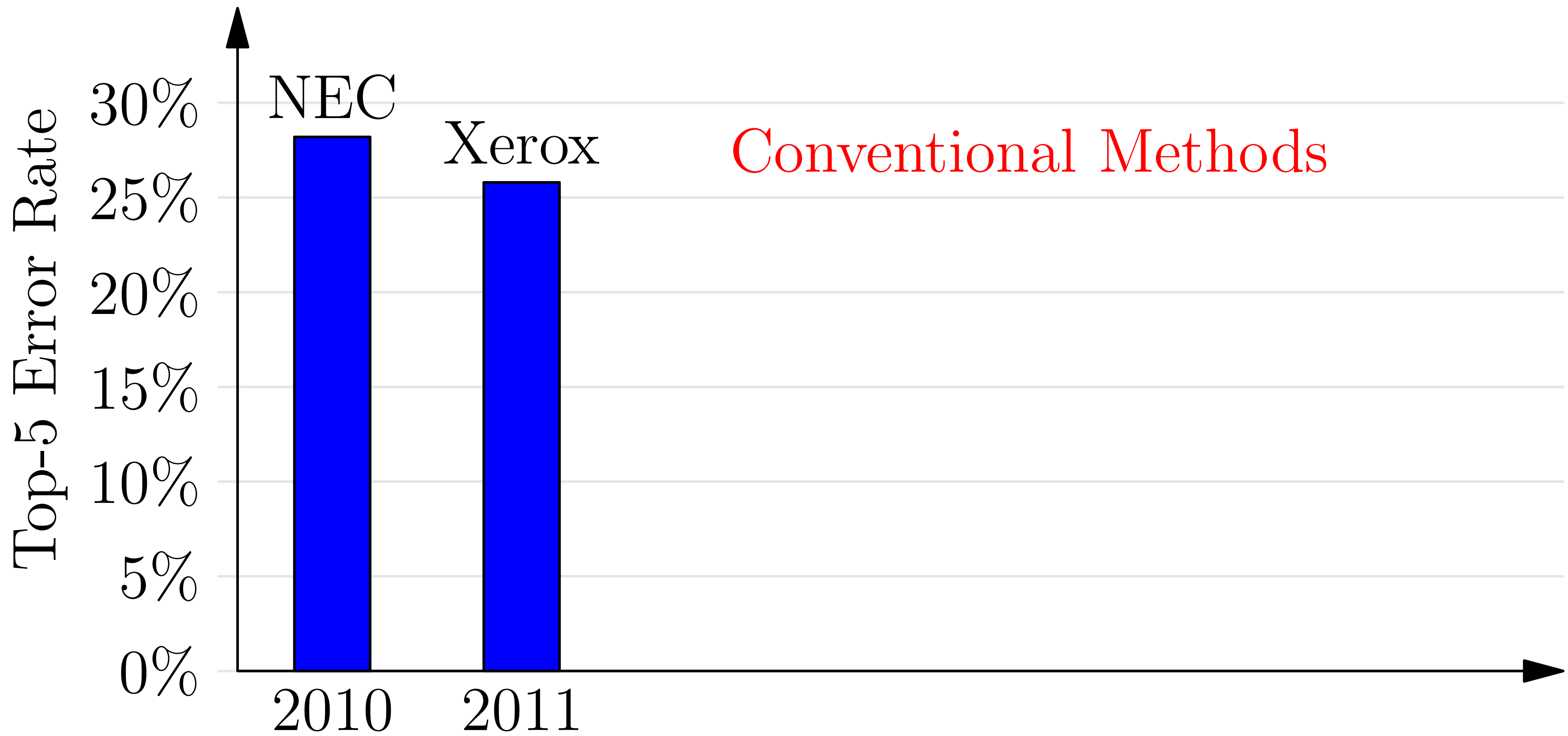

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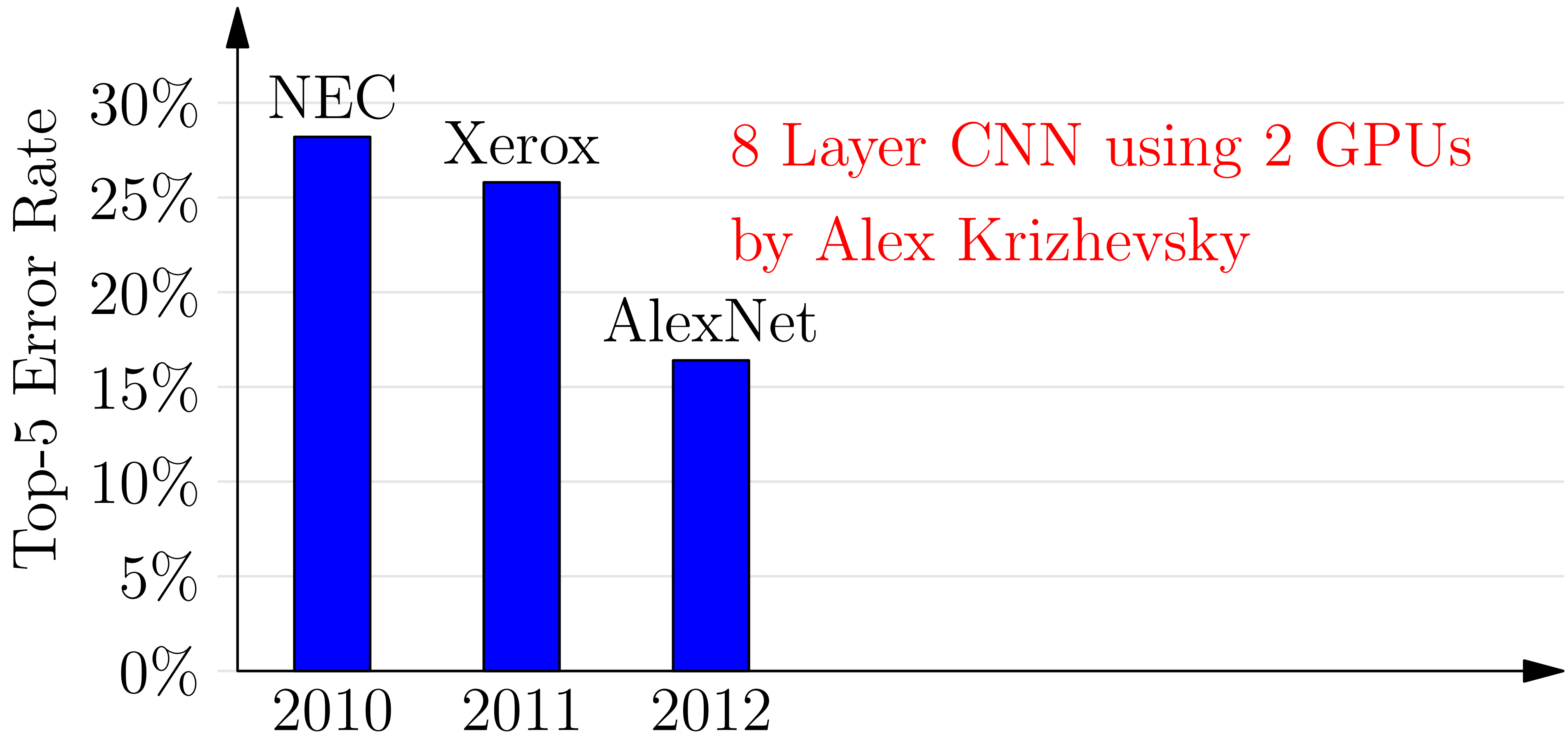

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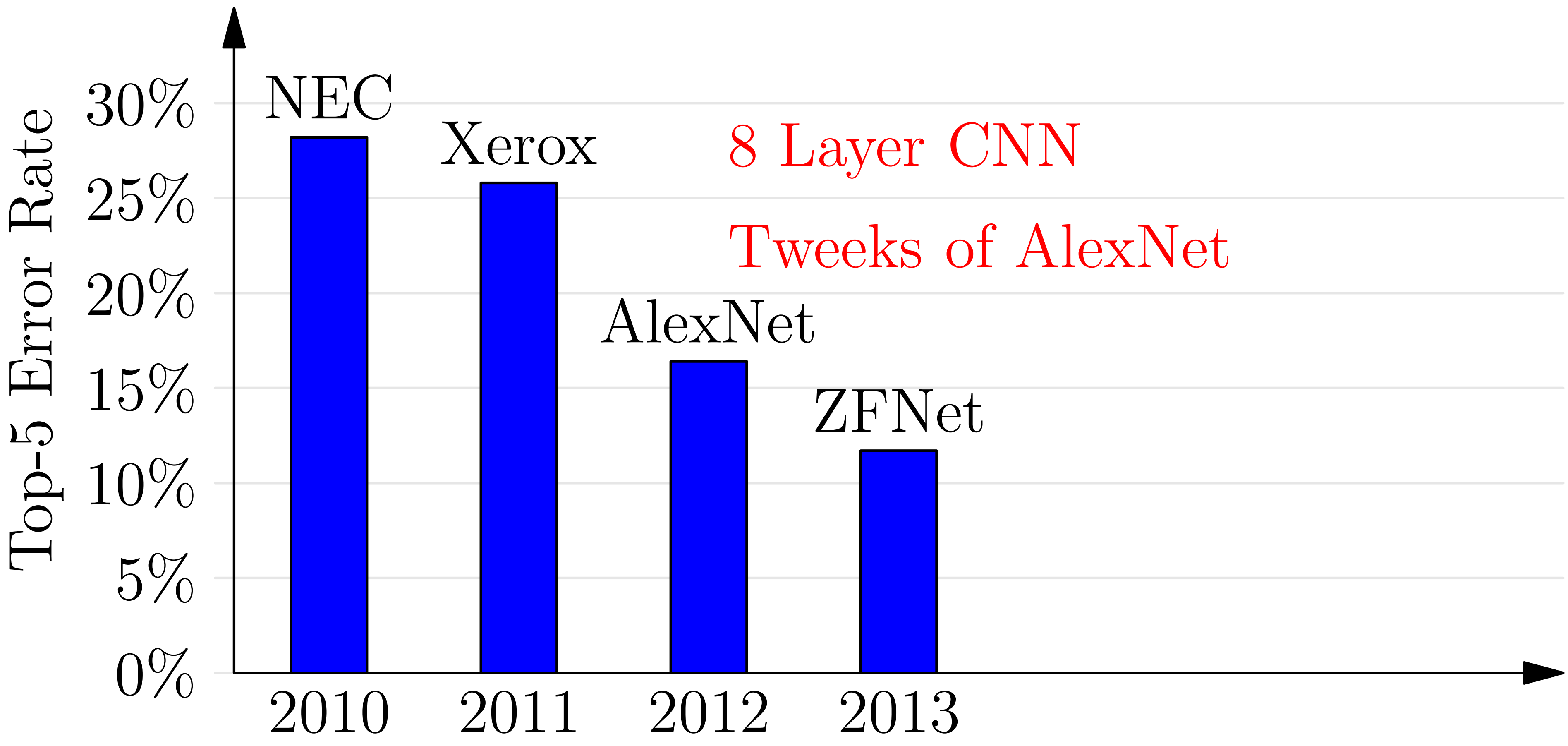

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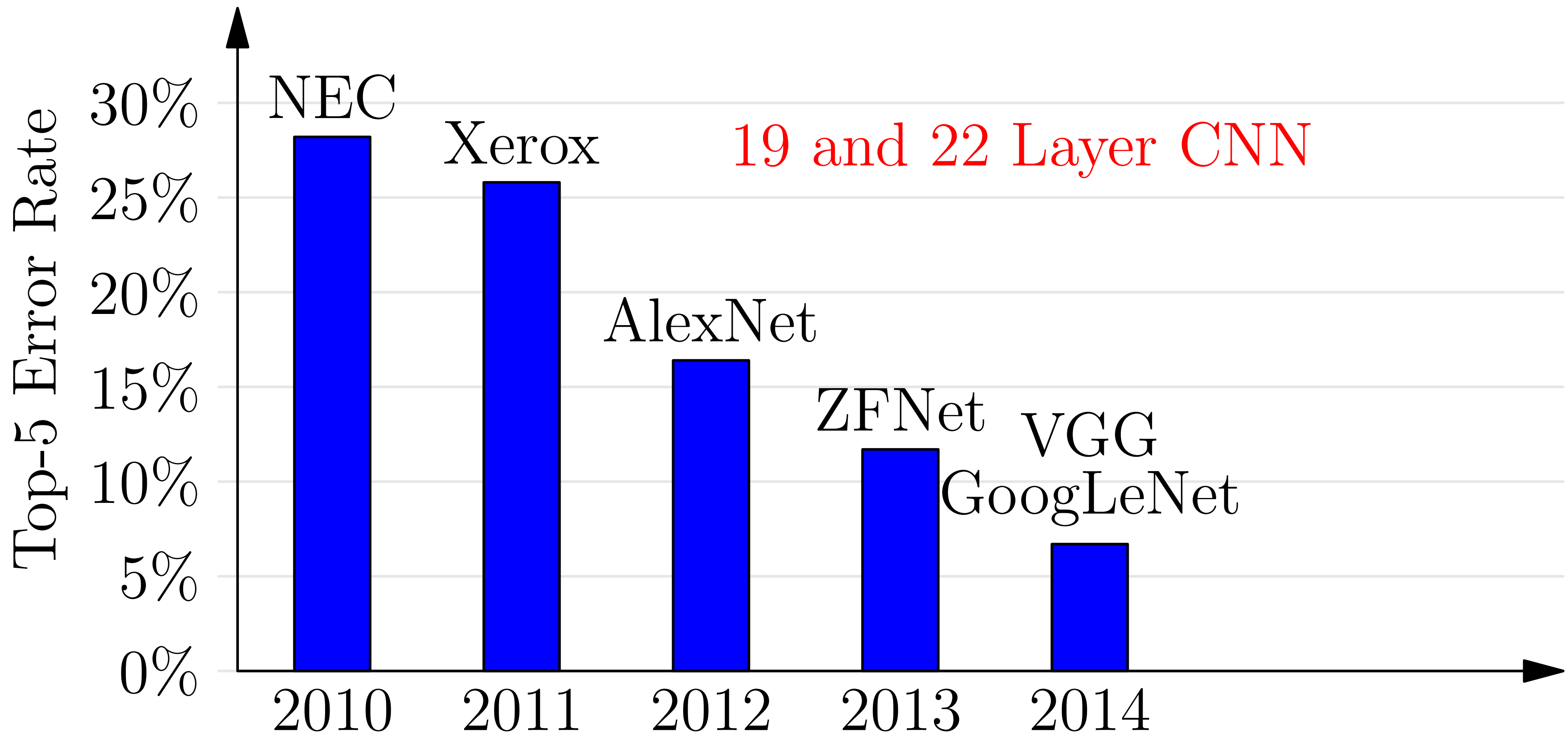

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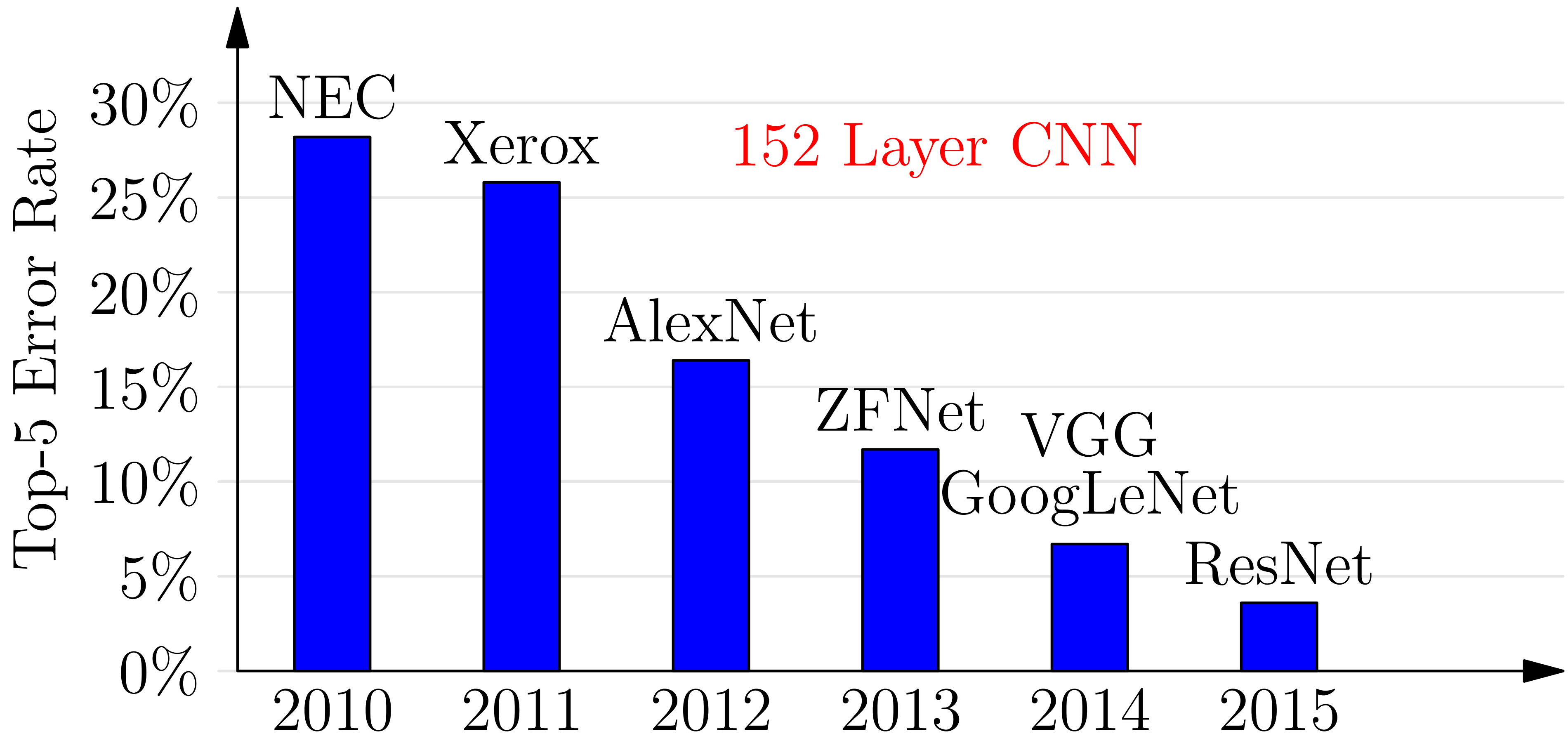

Dataset Size Matters

- The surprising result was not only did the model do extremely on the ImageNet data
- But, the network could handle real world data
- If you train on a large enough dataset we discovered that machine learning actually worked in the real world!
- Better models would give small improvements in results, but data really matters!

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What was Different About AlexNet?

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- To train on bonkers amount of data you need a lot of compute, and AlexNet introduced **GPUs**
- Traditionally, we had to compute gradients by hand, making the design of large networks cumbersome
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The Unreasonable Effectiveness of SGD

- An extraordinarily successful learning strategy is to design a standard network

- This scales well (linearly) to large networks and large datasets

- When you overfit, you just turn on some regularisation

- Sometimes gradient descent would struggle, but usually changing the optimiser to ADAM or fine-tuning some hyper-parameters would do the trick

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Computational Frameworks

- The Machine Learning community is rubbish when it comes to organising conferences and journals
- But when it comes to developing computational frameworks it is brilliant
- It rapidly developed, torch (Lua), TensorFlow, Keras, Pytorch, JAX, etc.
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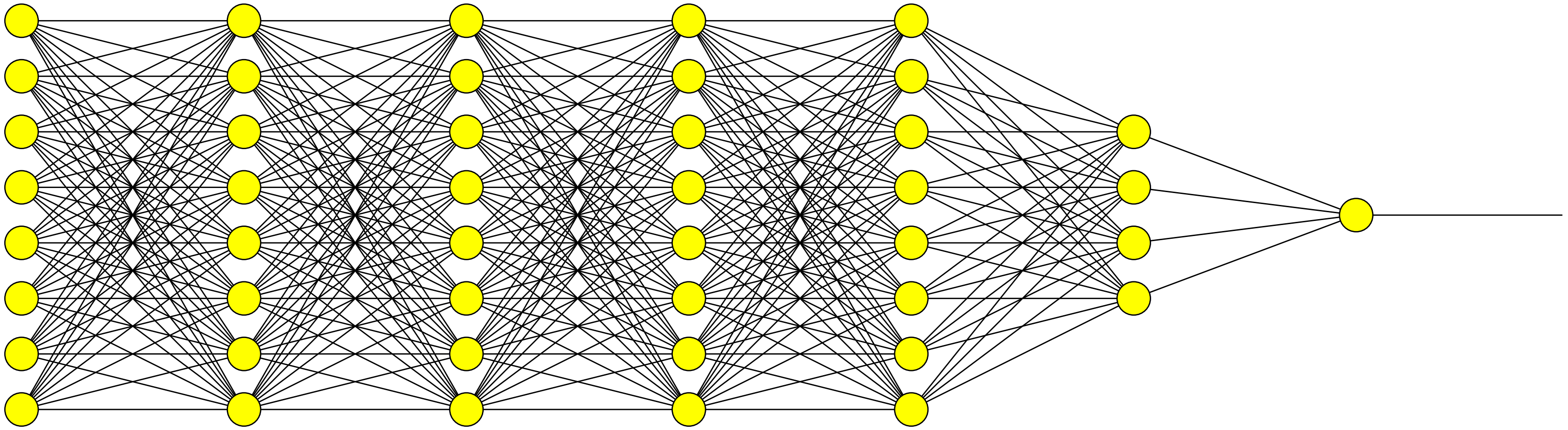

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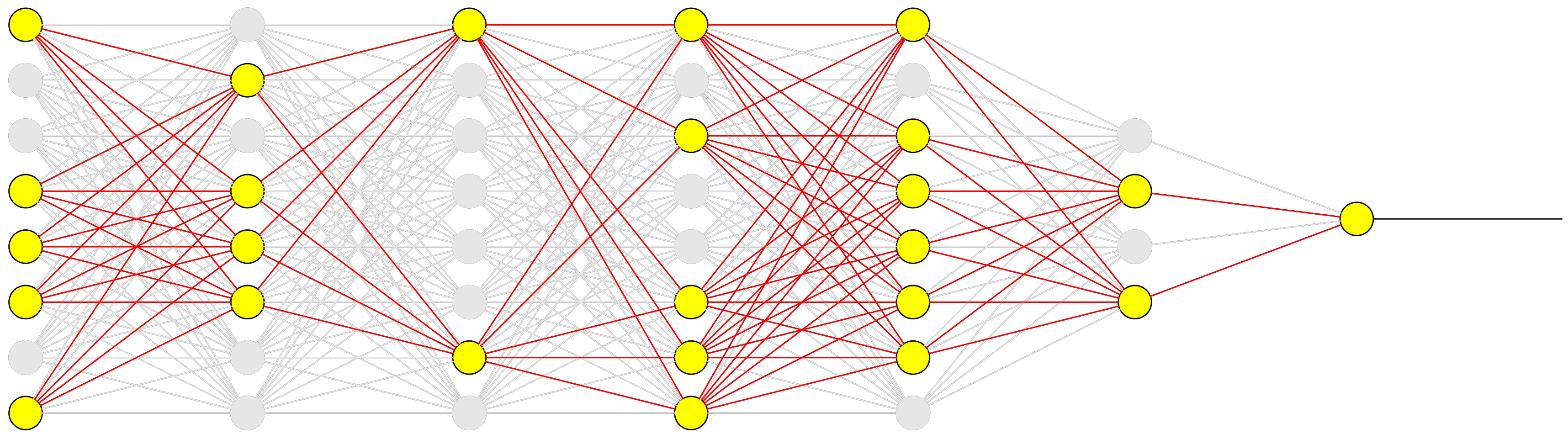

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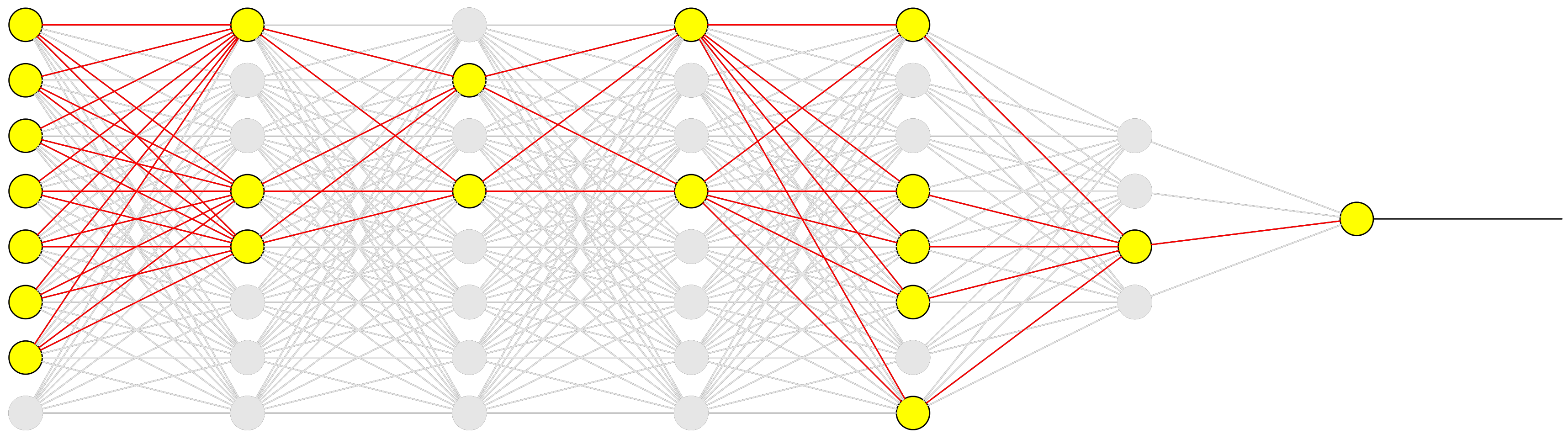

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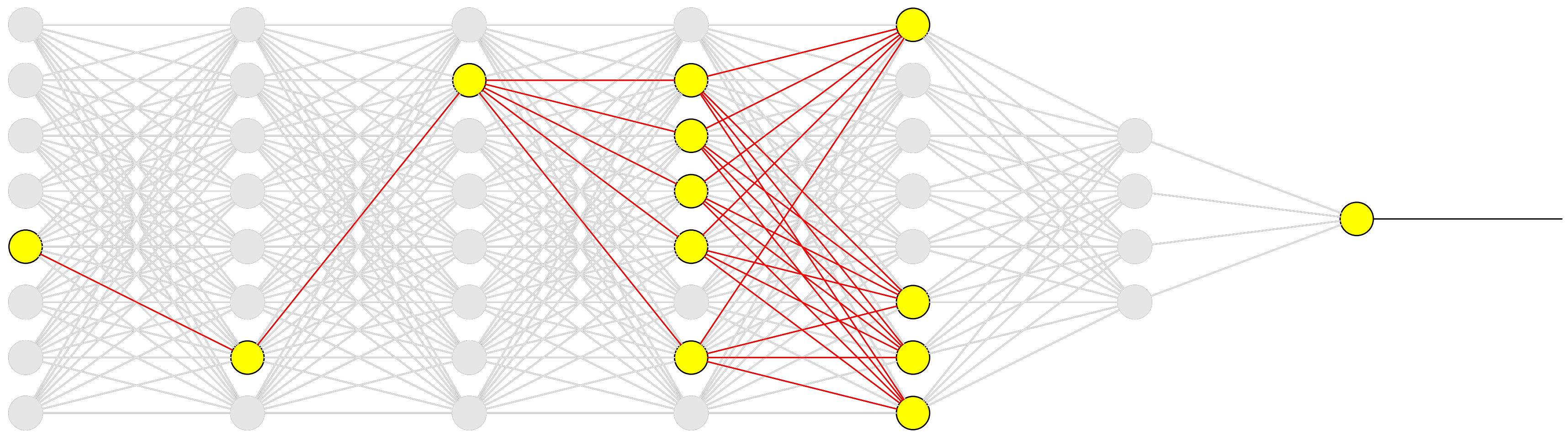

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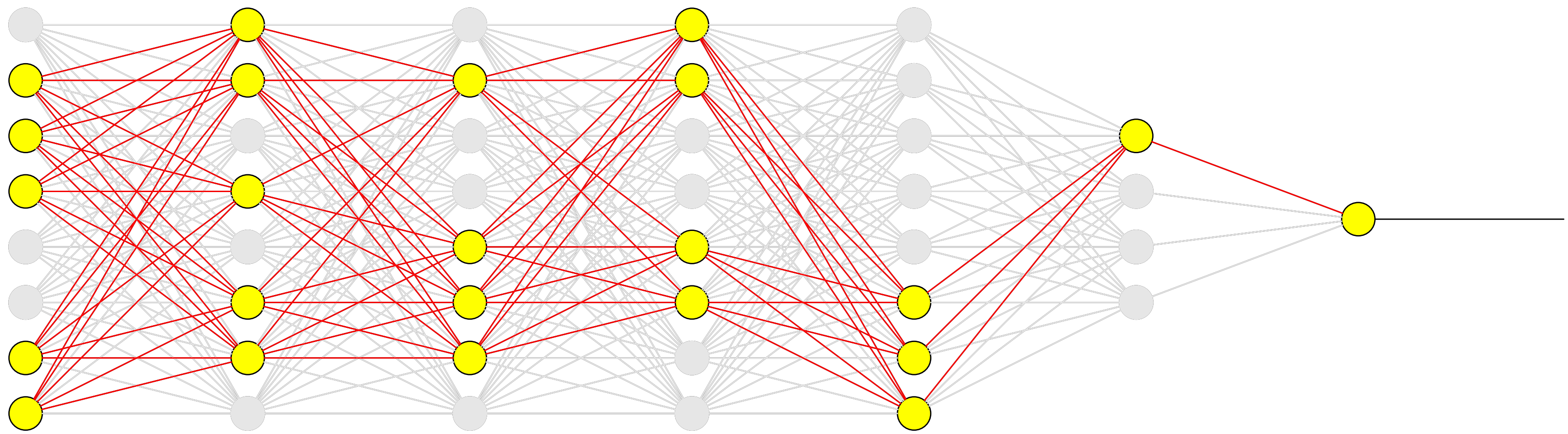

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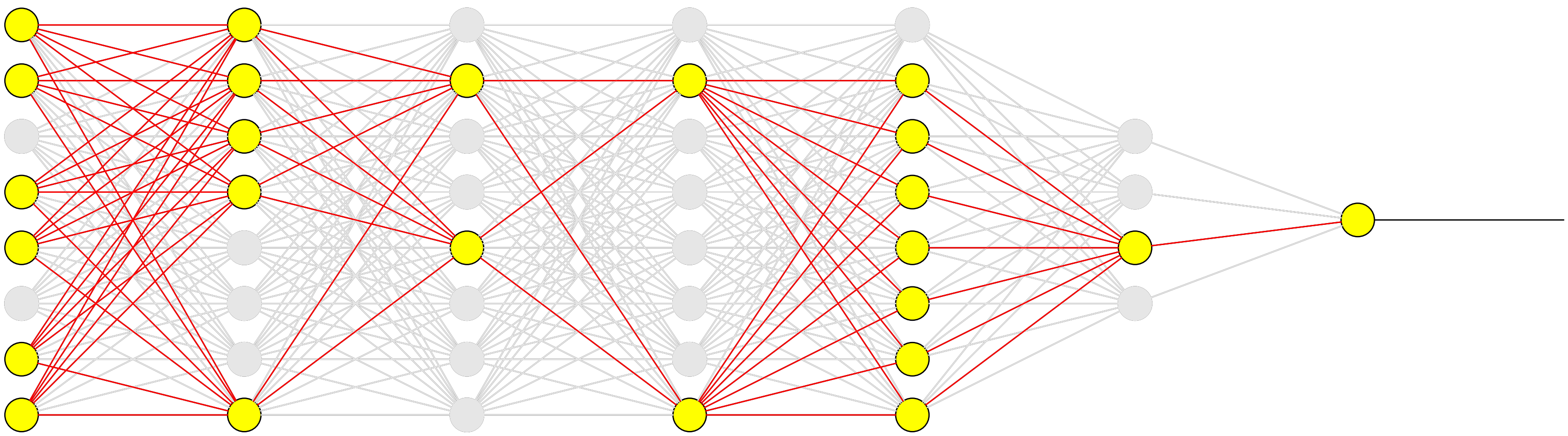

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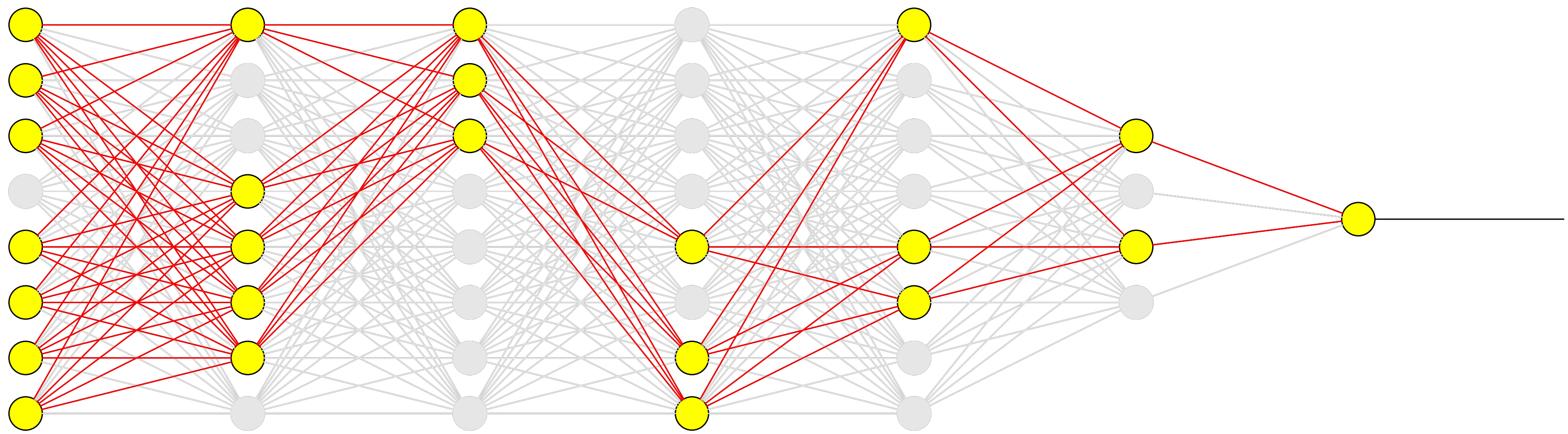

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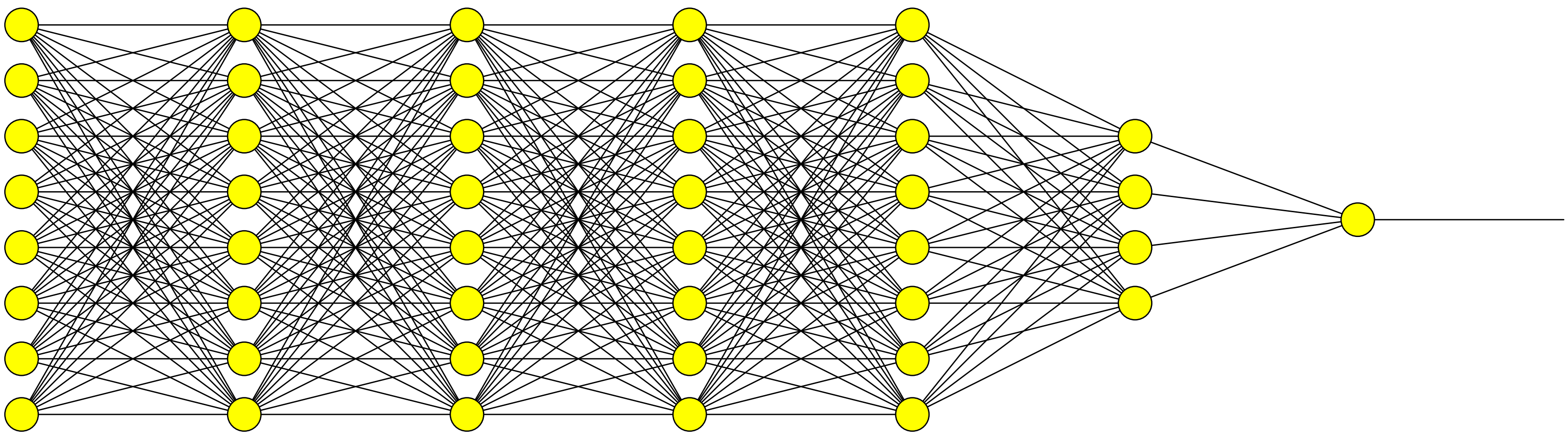

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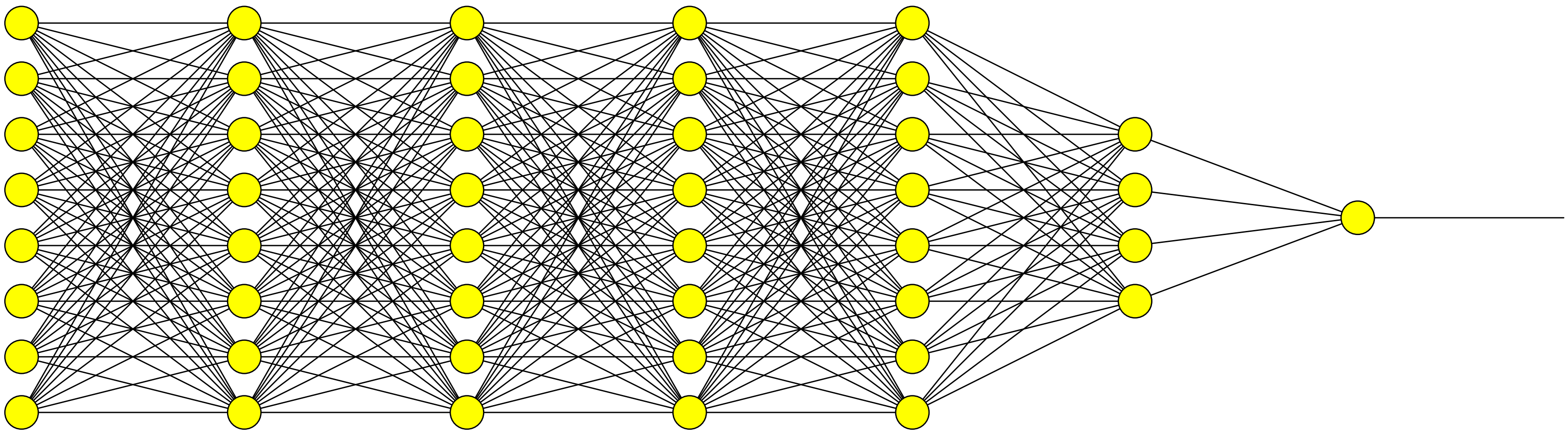

Acts as a strong form of regularisation

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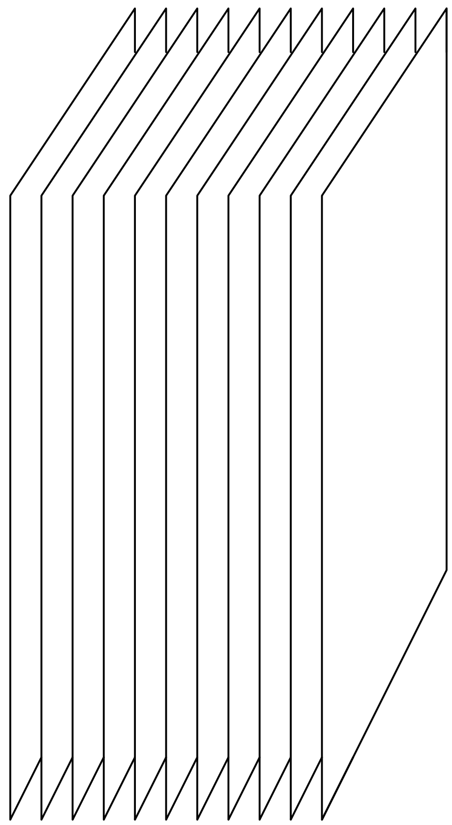

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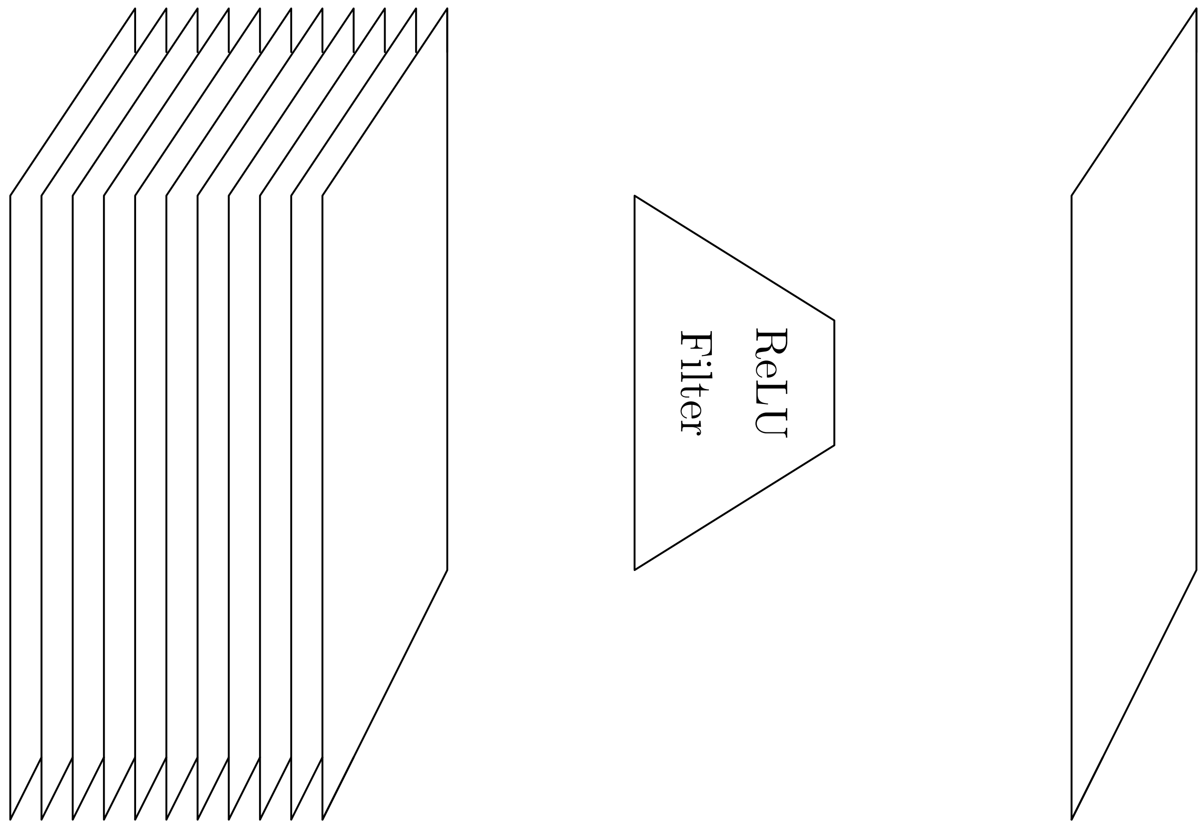

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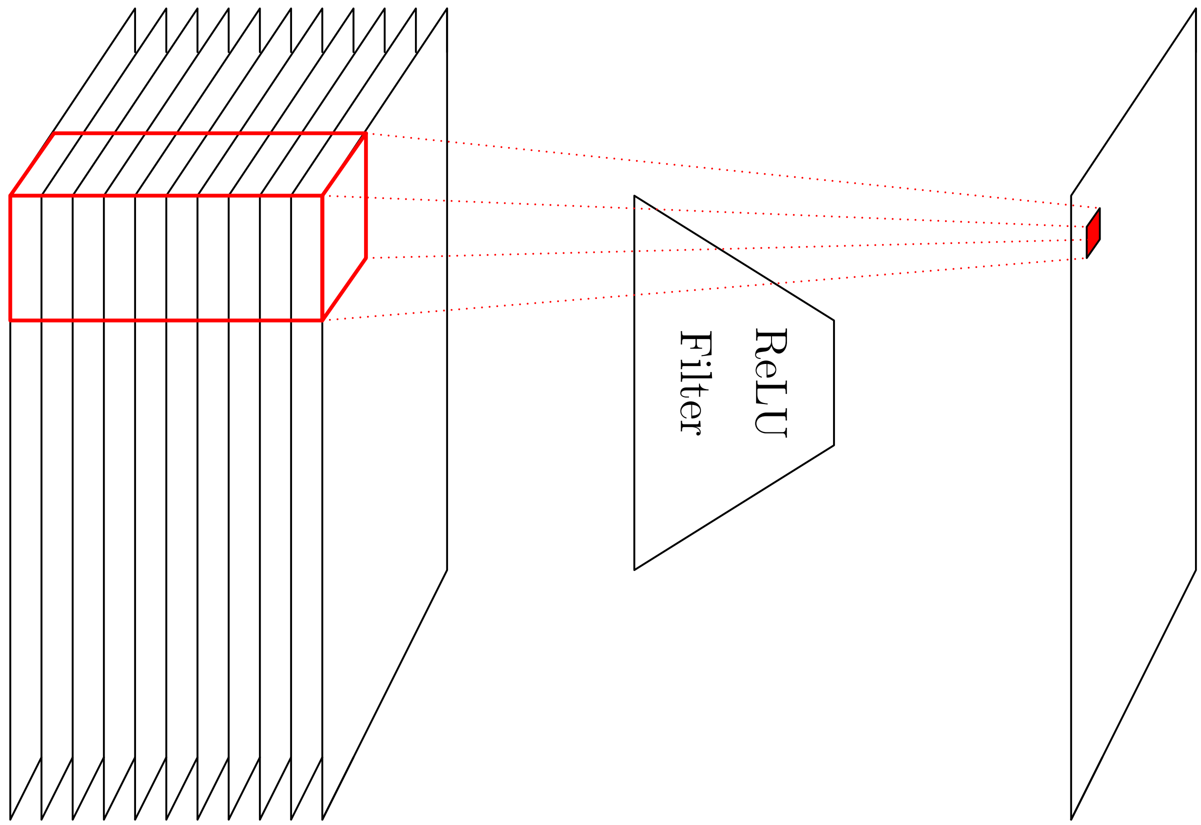

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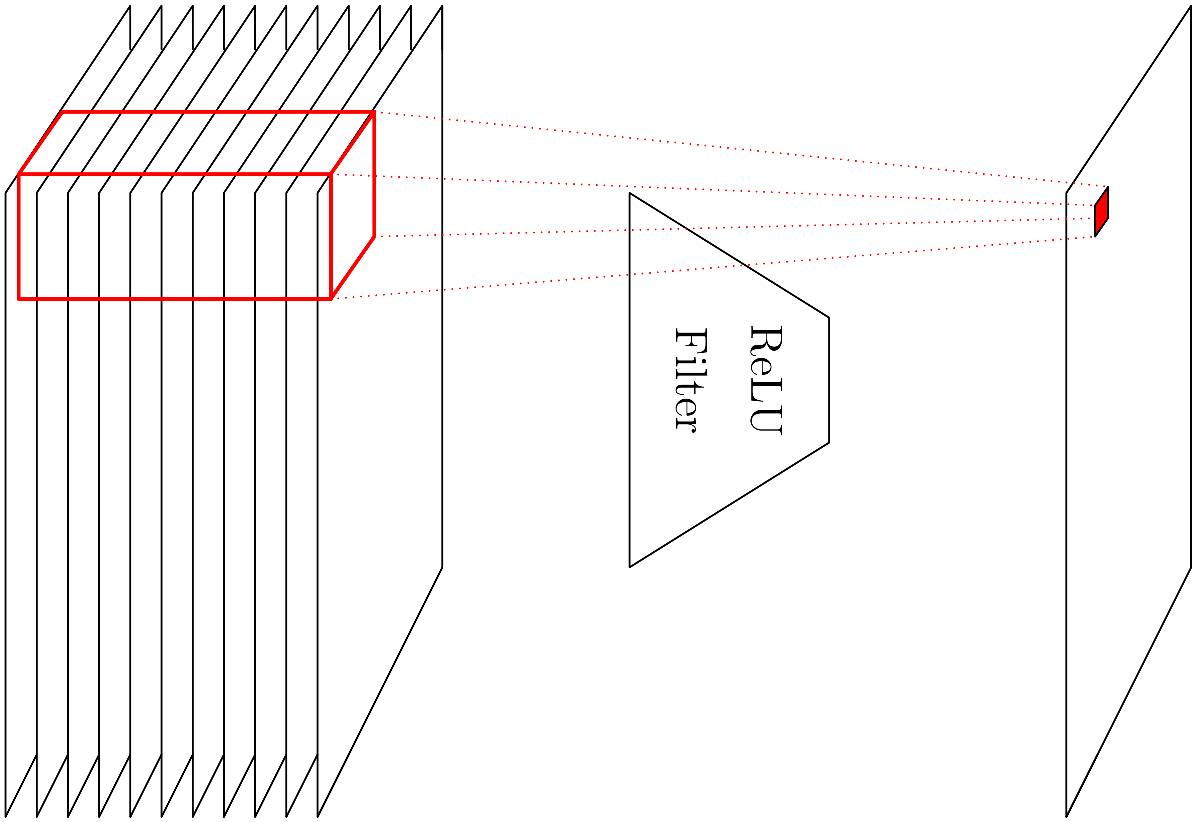

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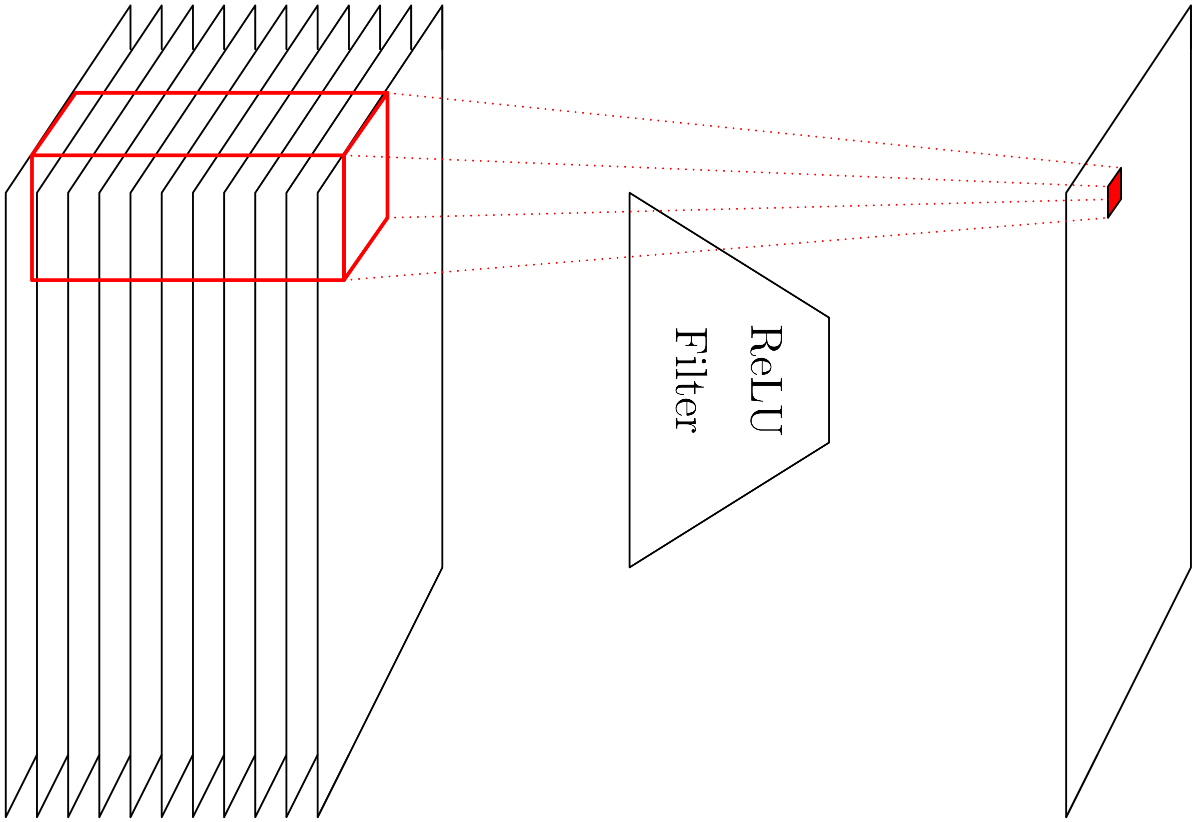

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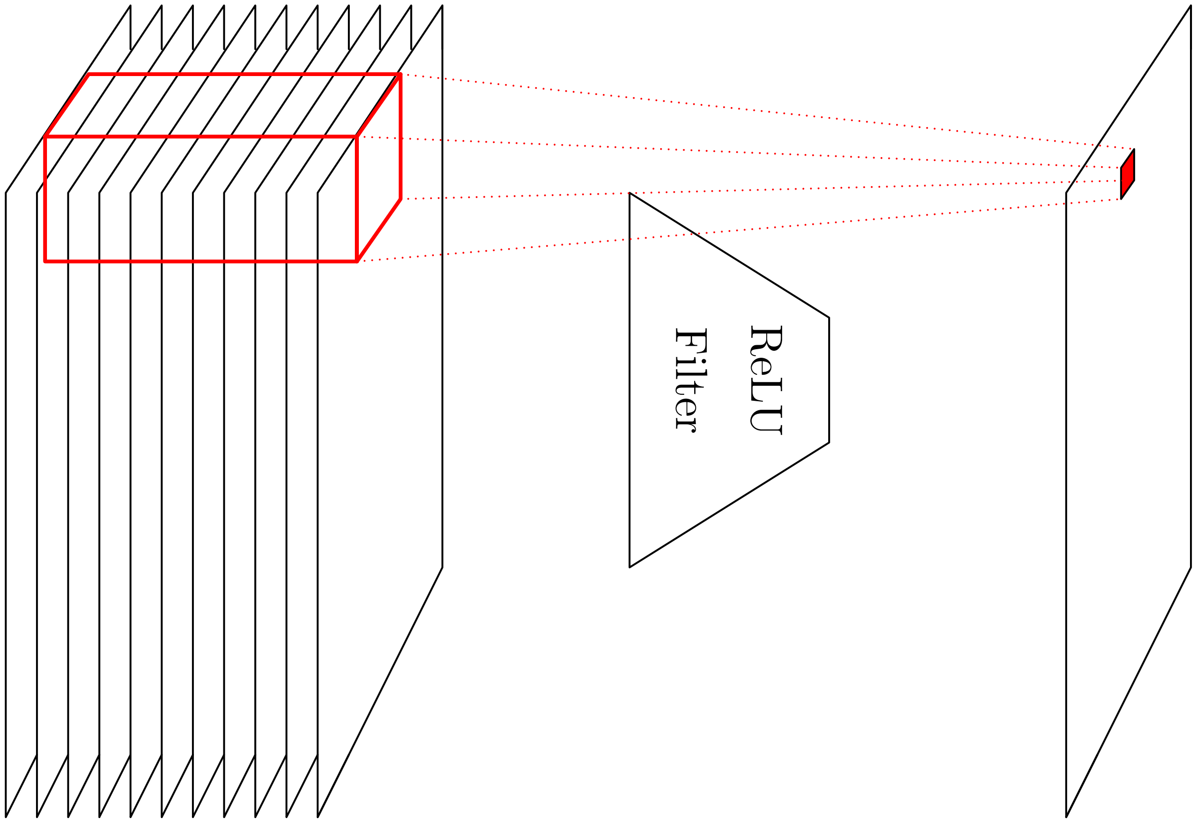

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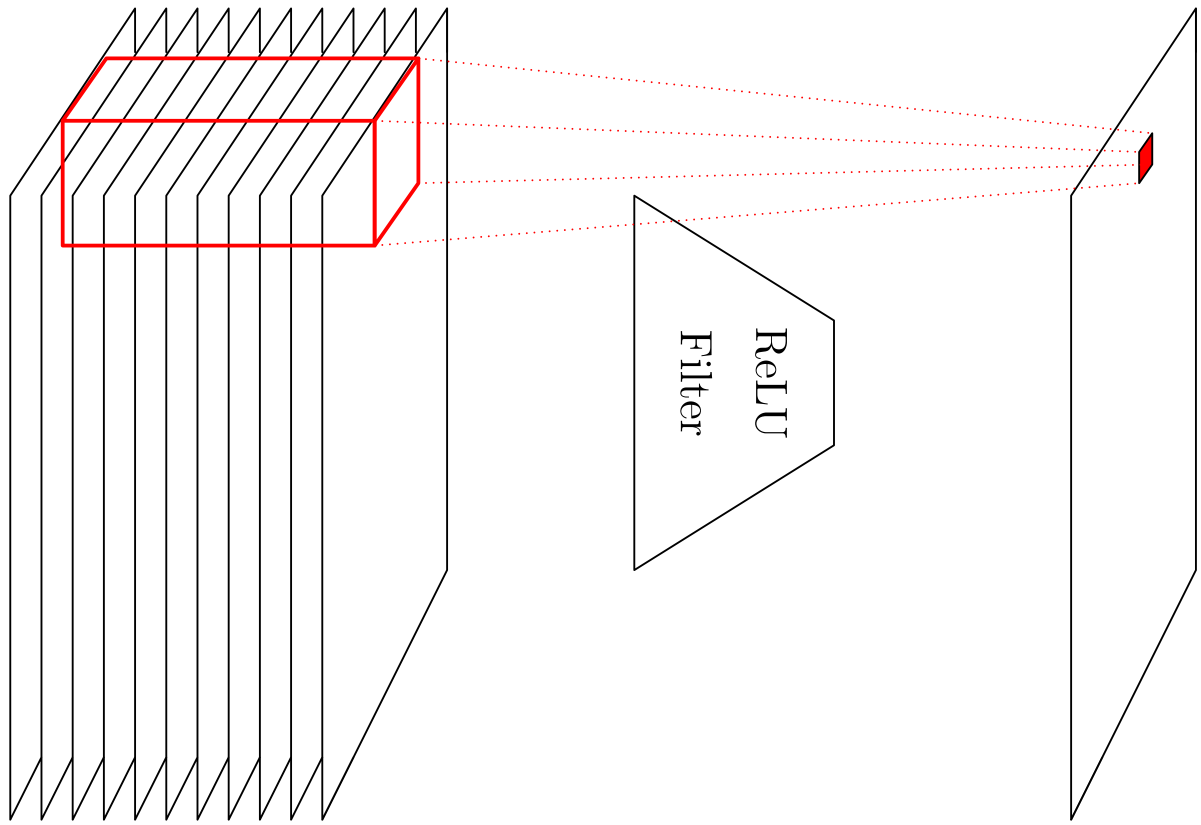

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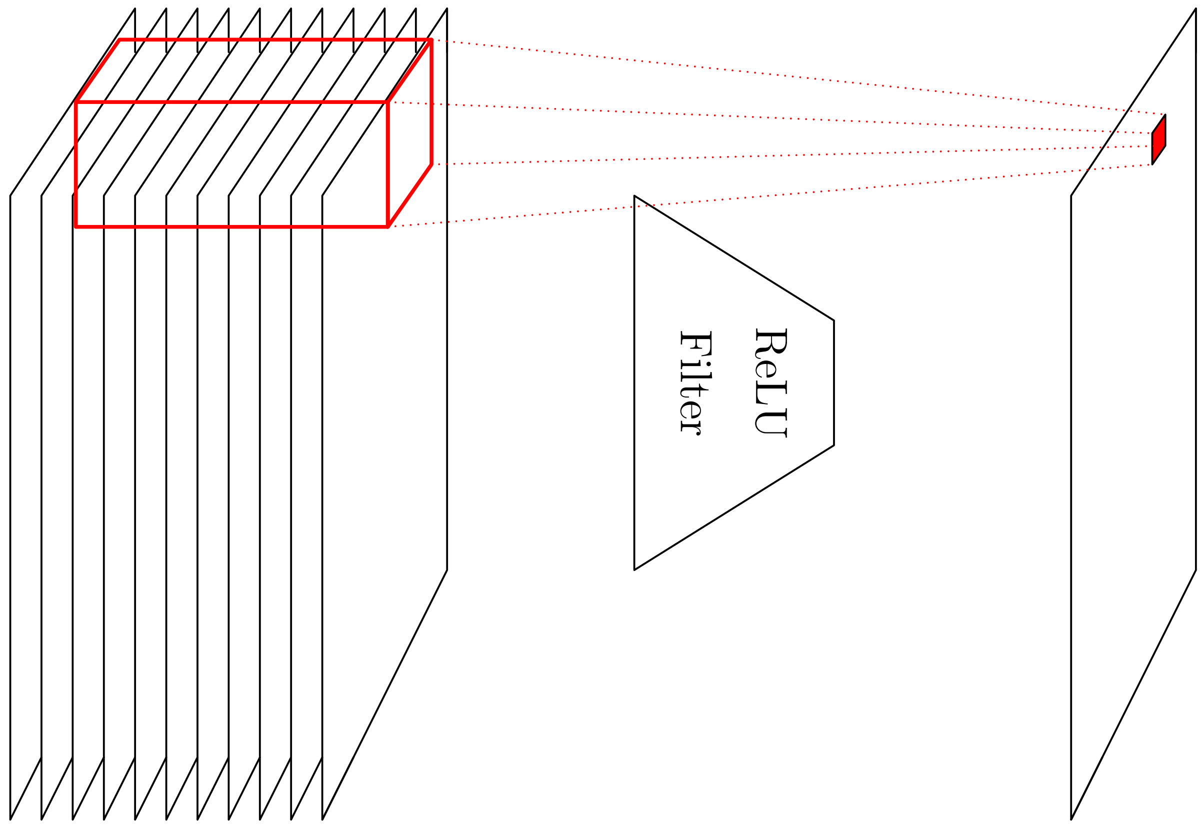

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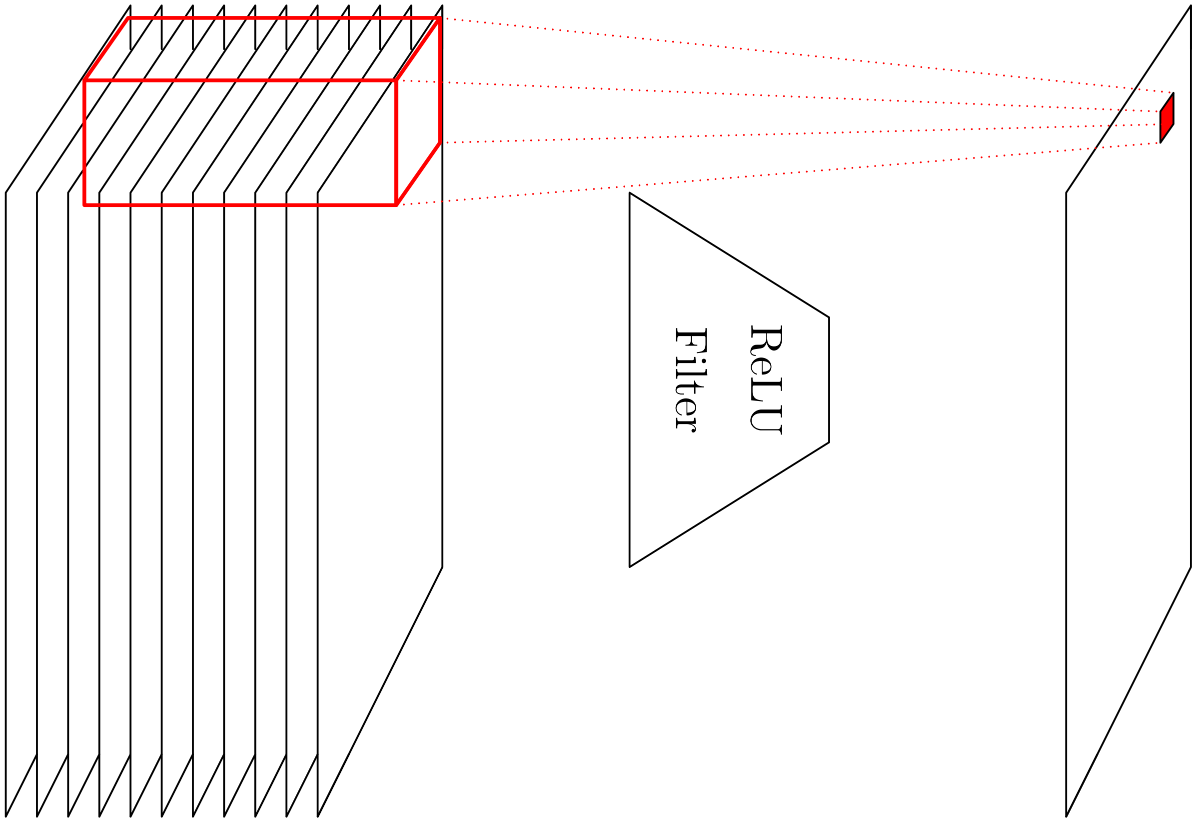

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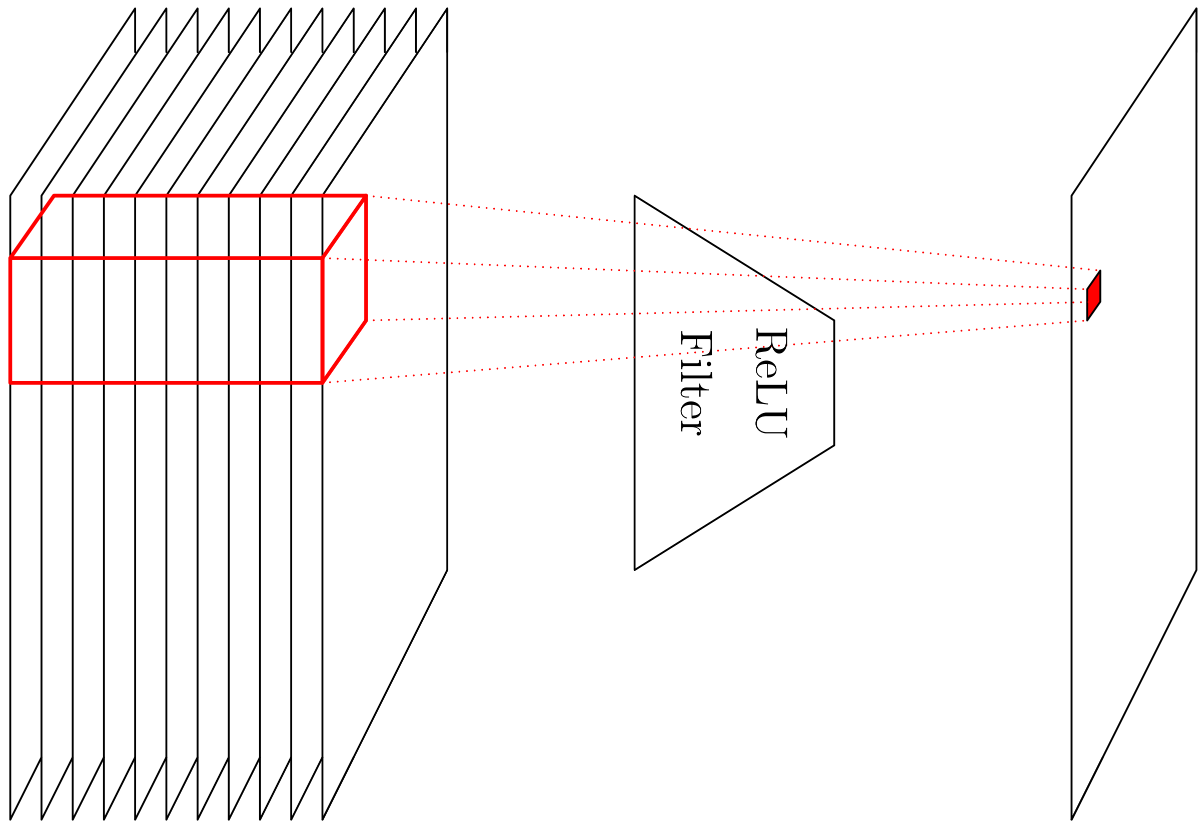

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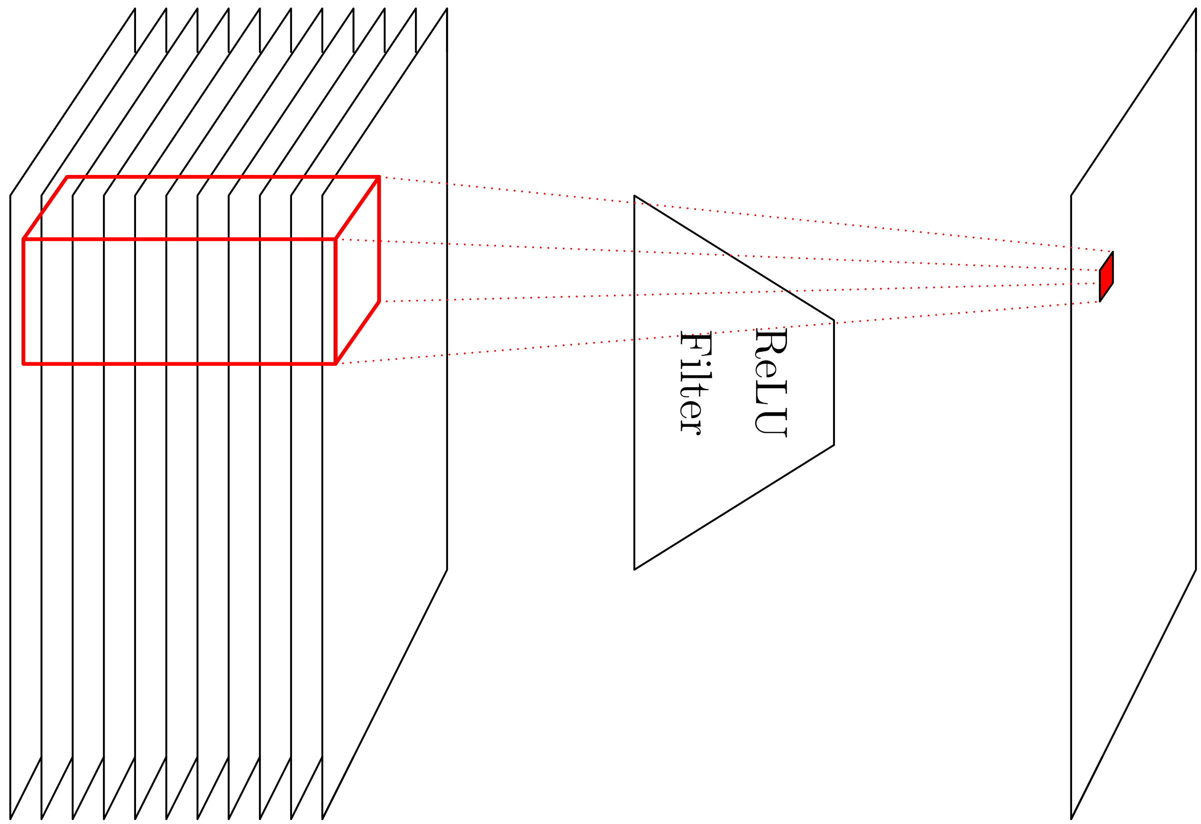

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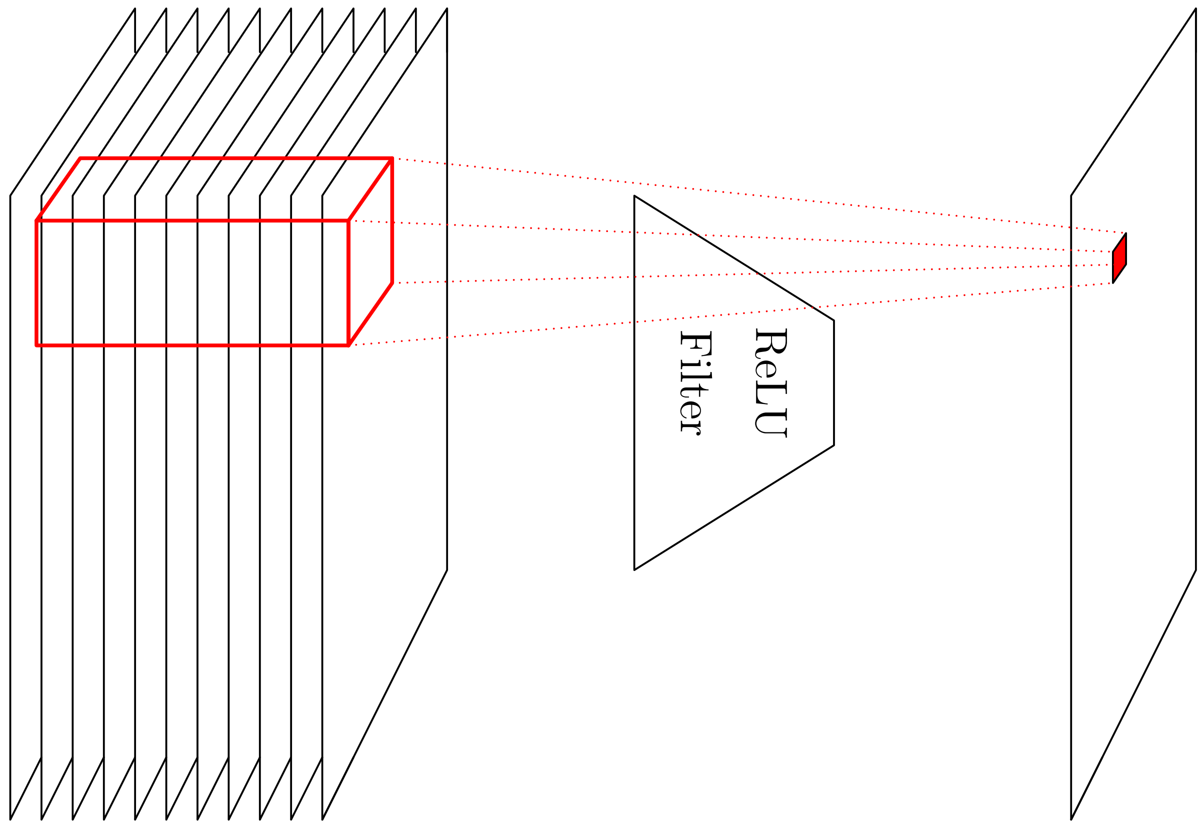

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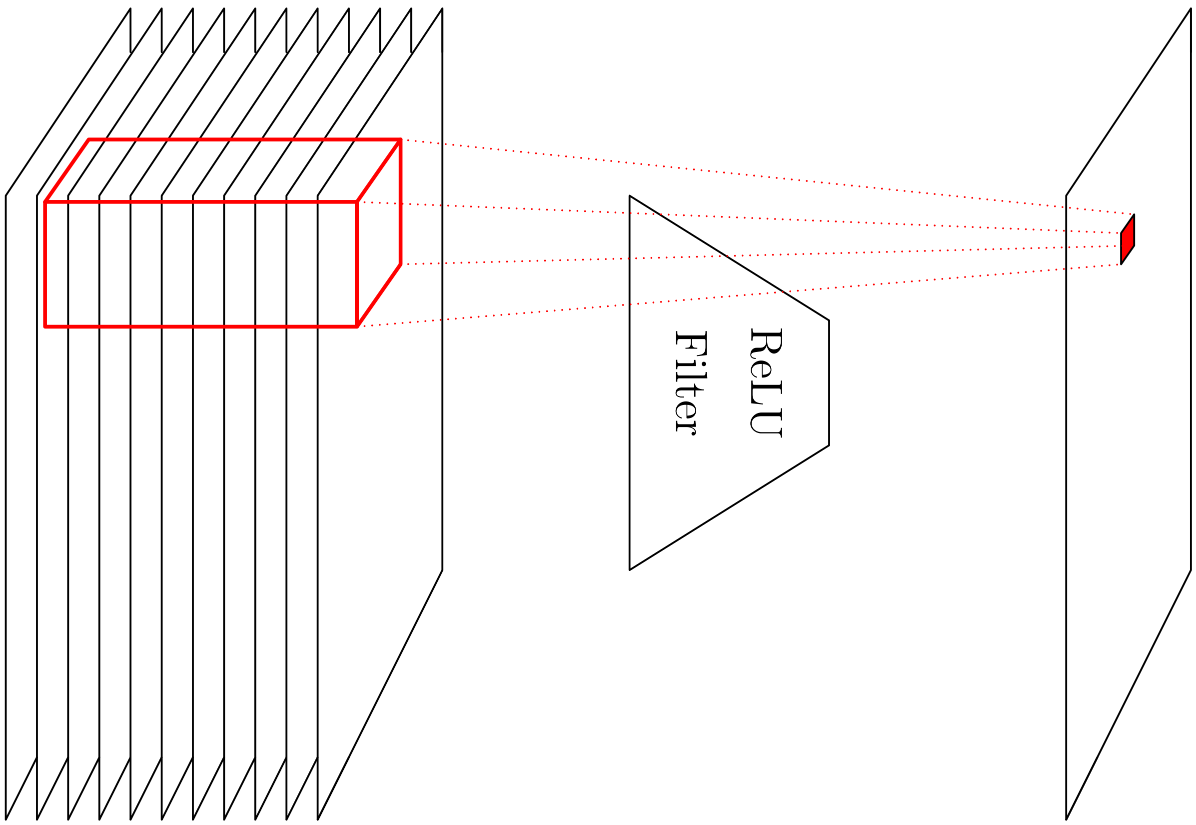

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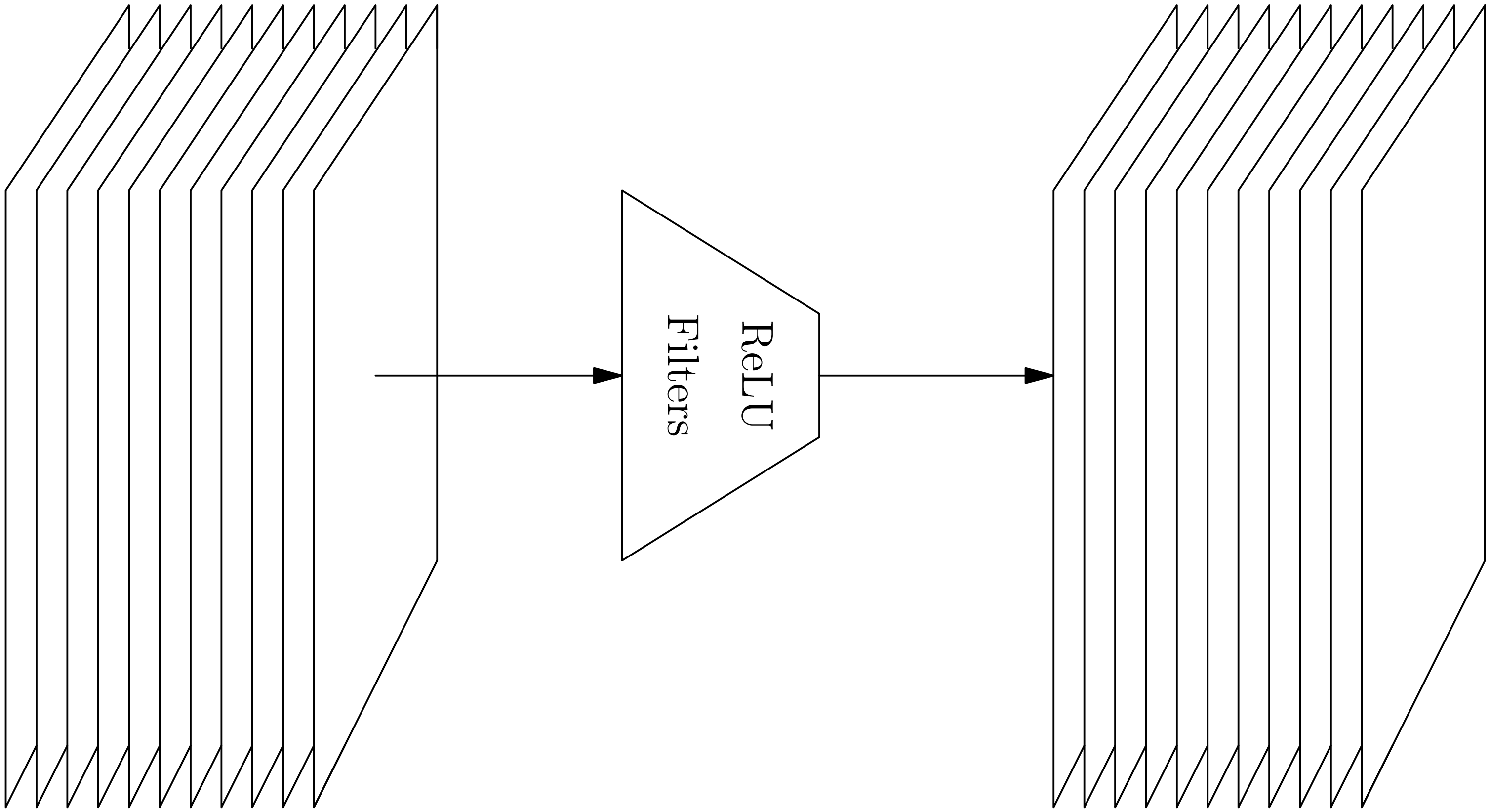

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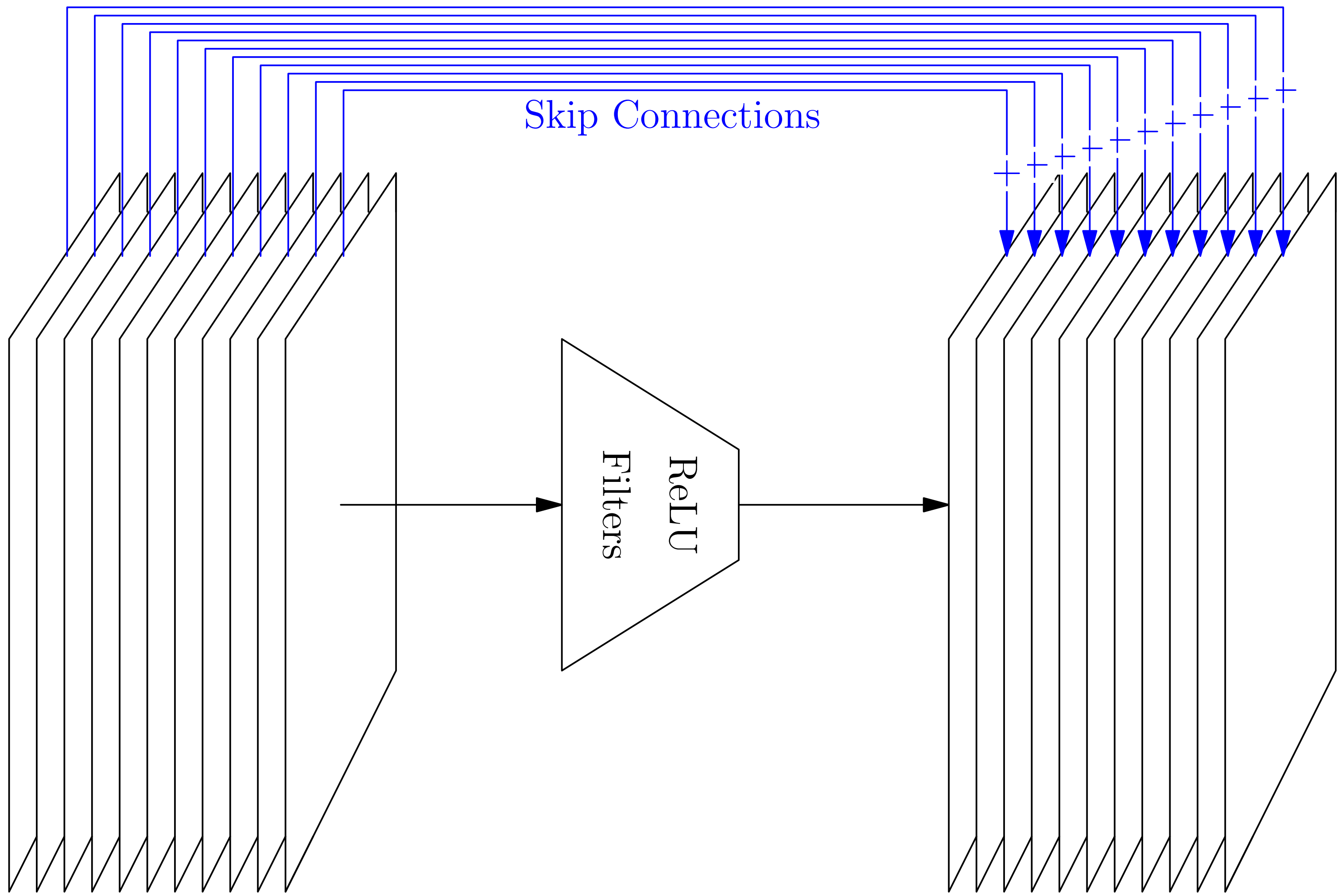

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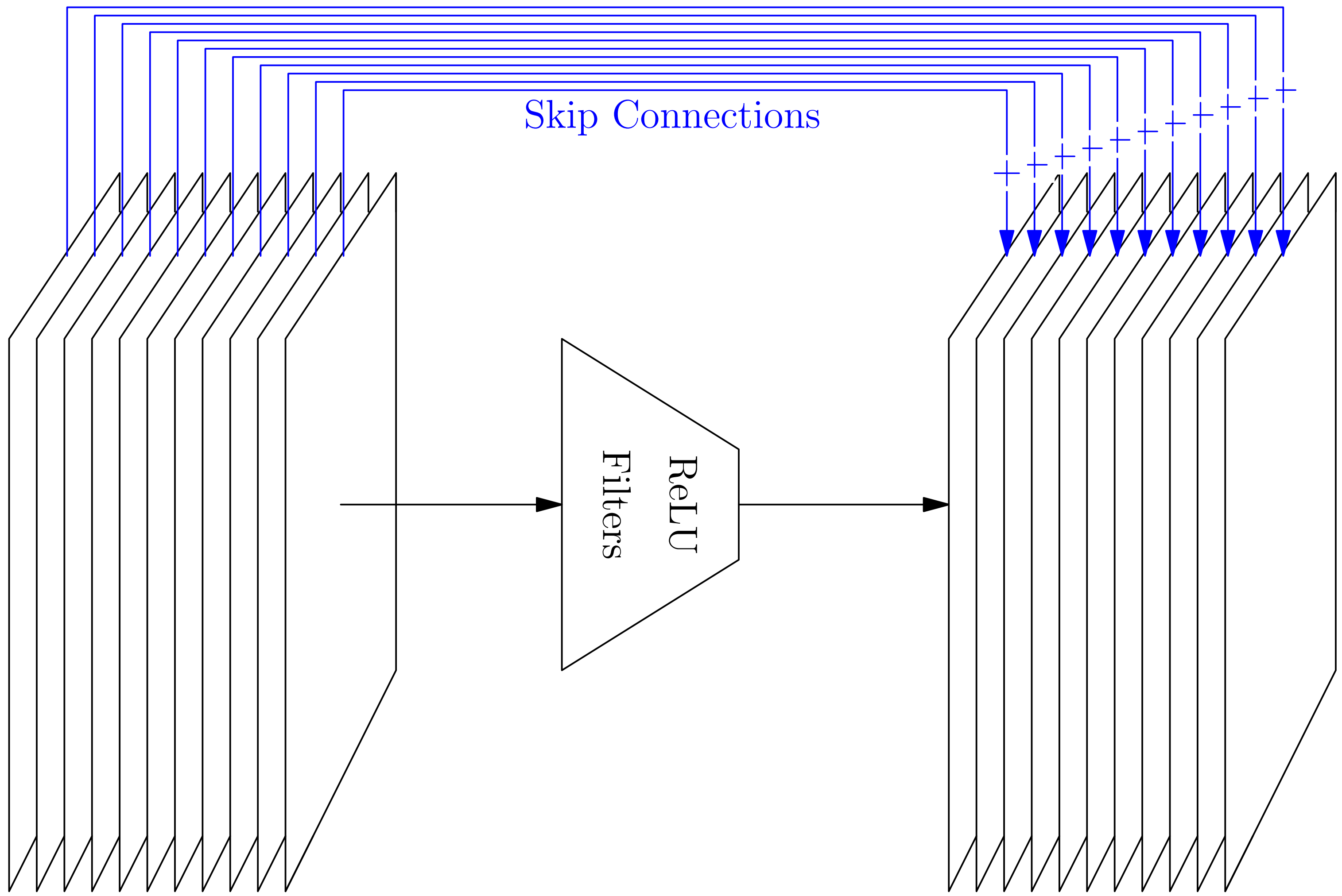
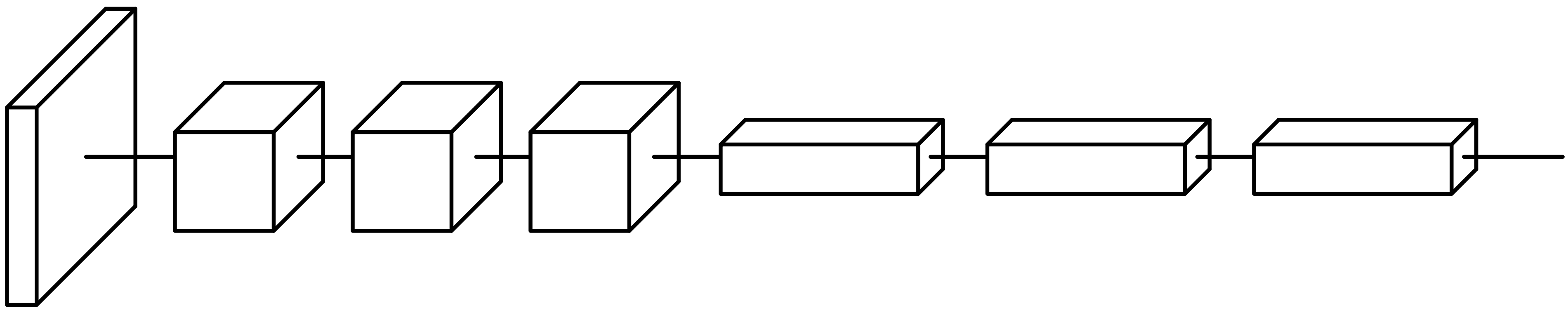


Breaks symmetry, improves signal and gradient vanishing

New Architectures: ResNet



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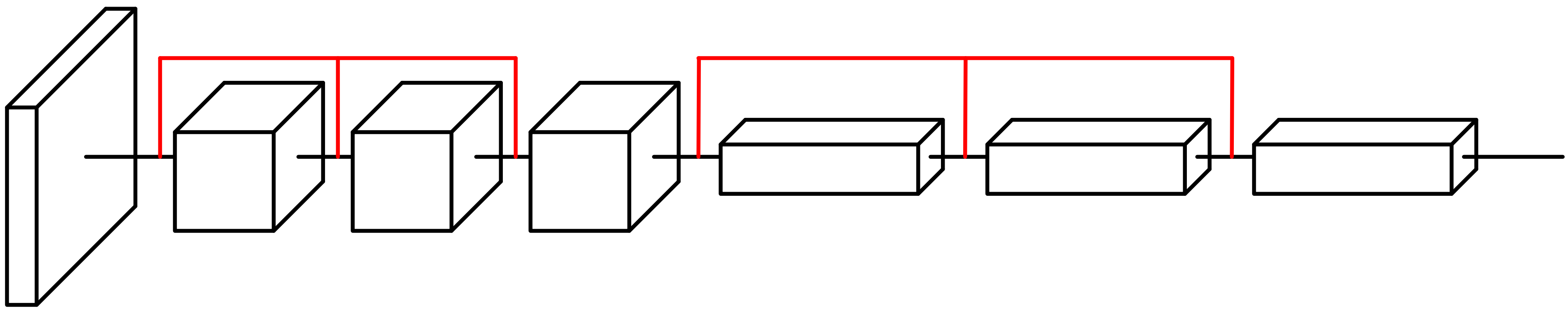

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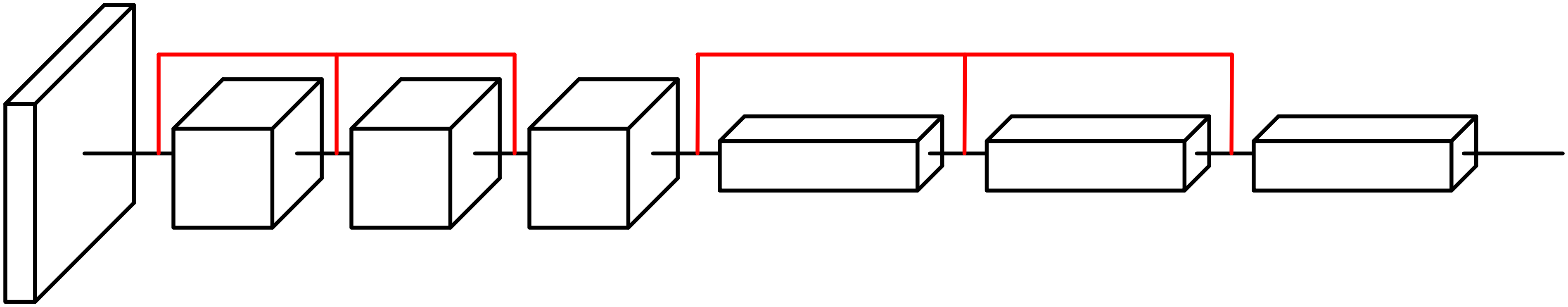
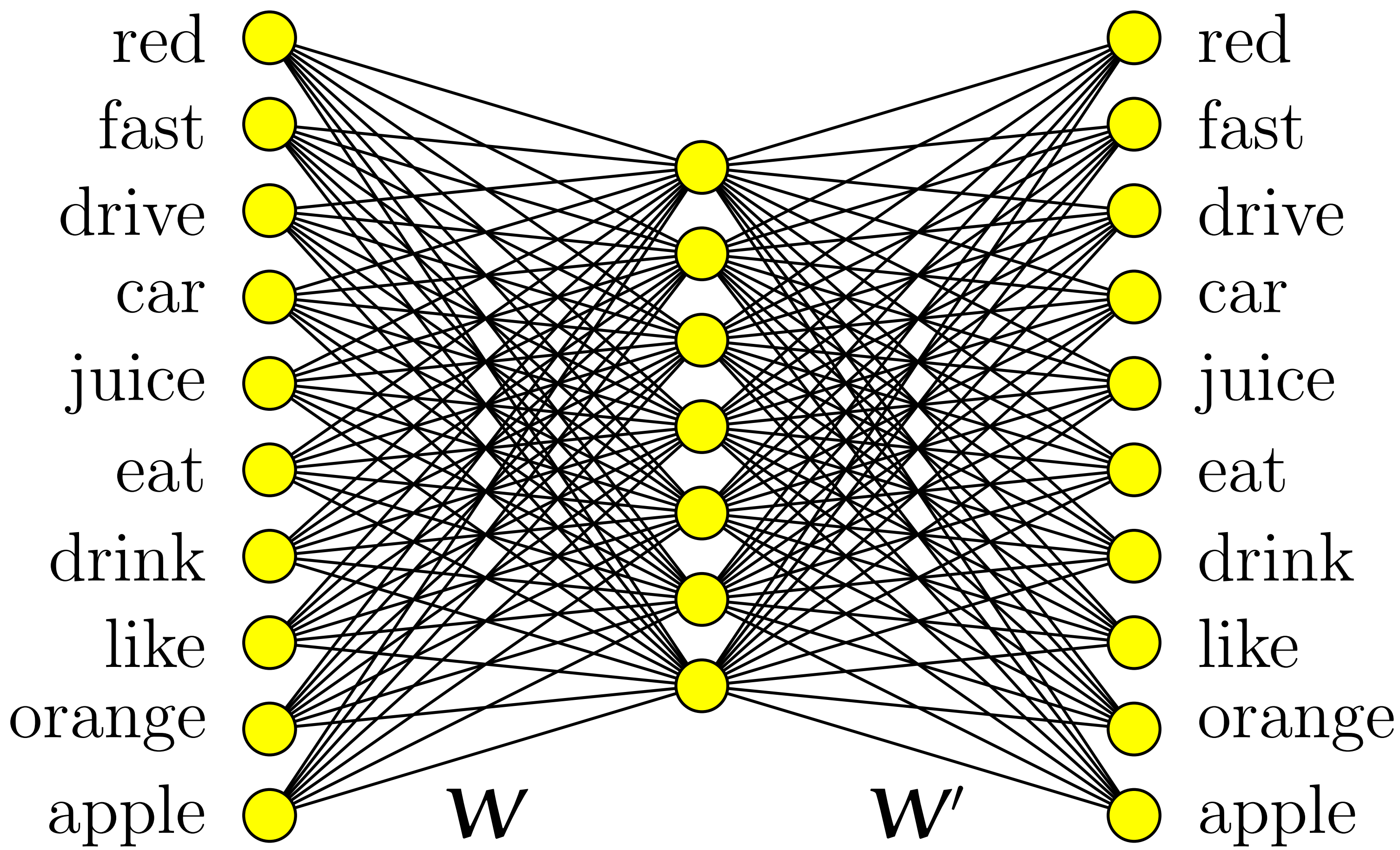


Used with batch normalisation, allowing very deep networks

New Ideas: Embeddings



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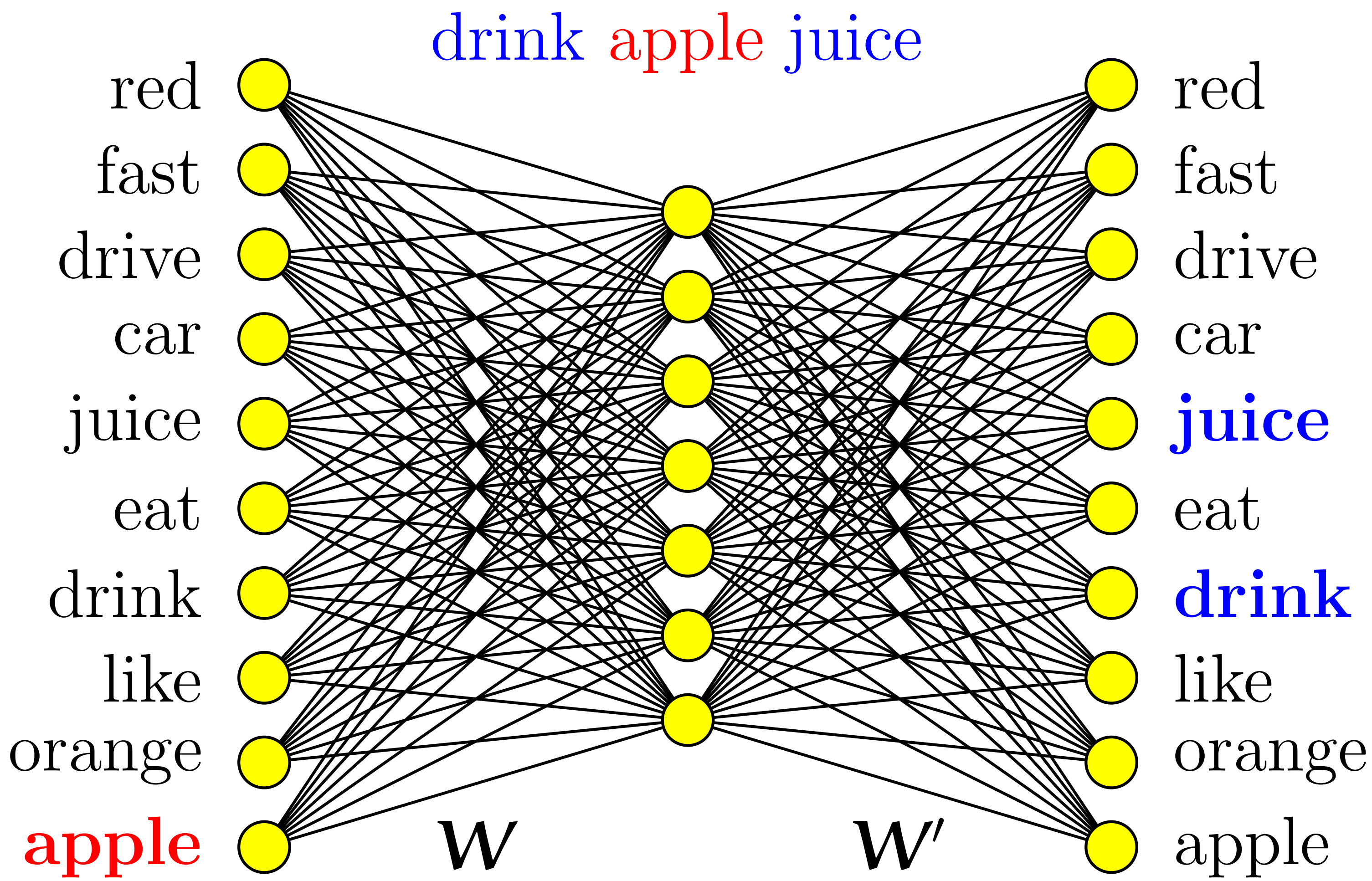

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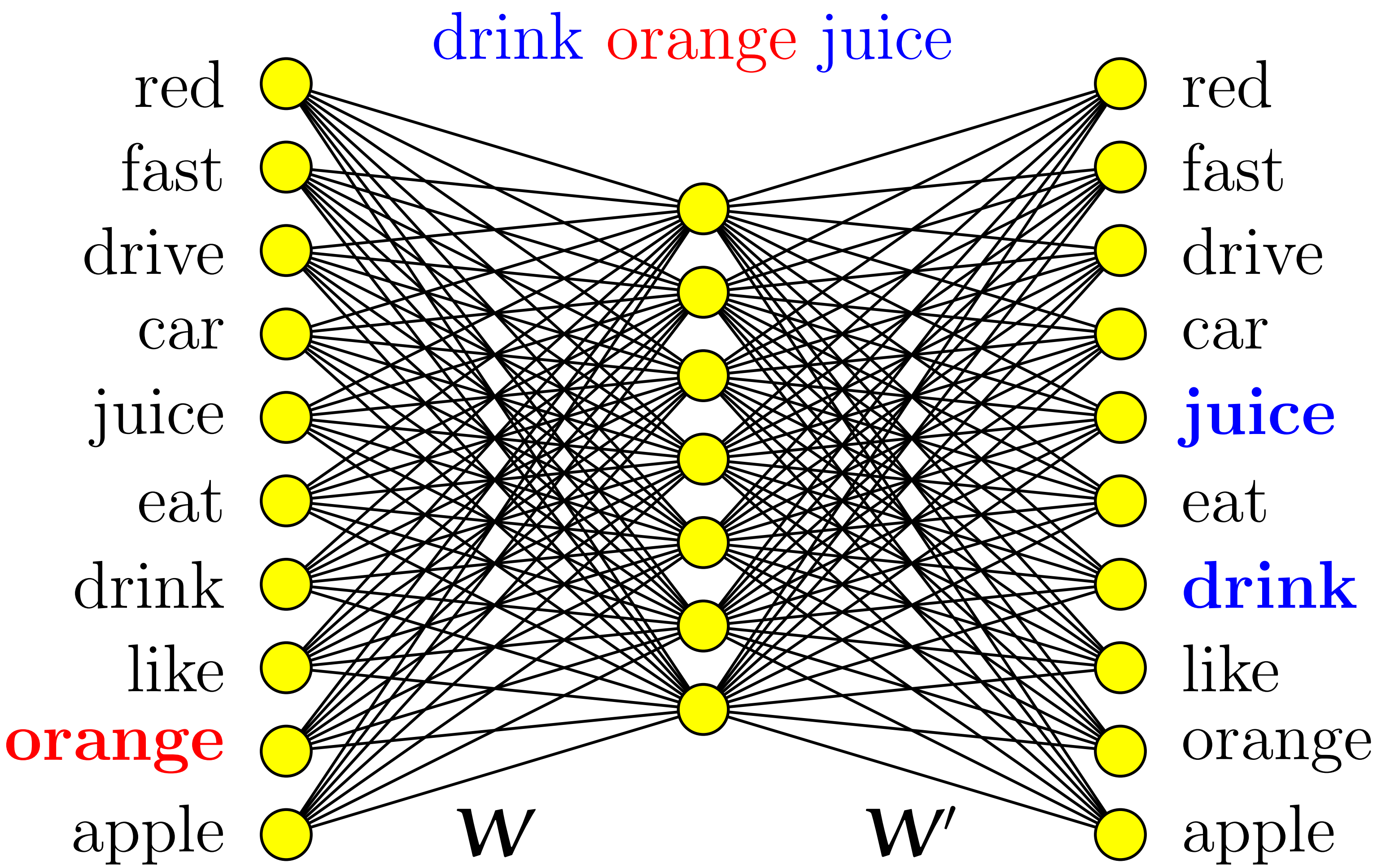

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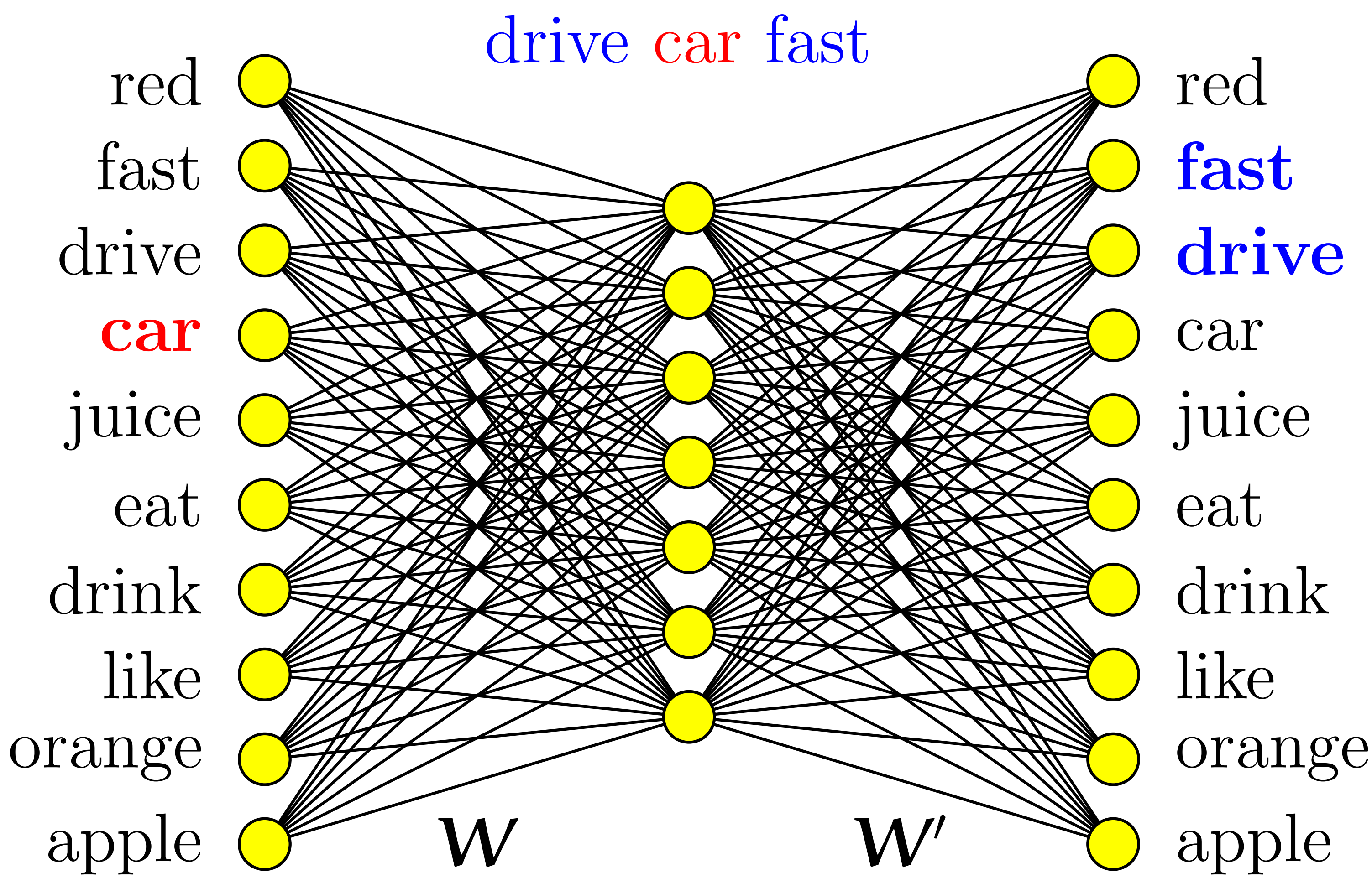

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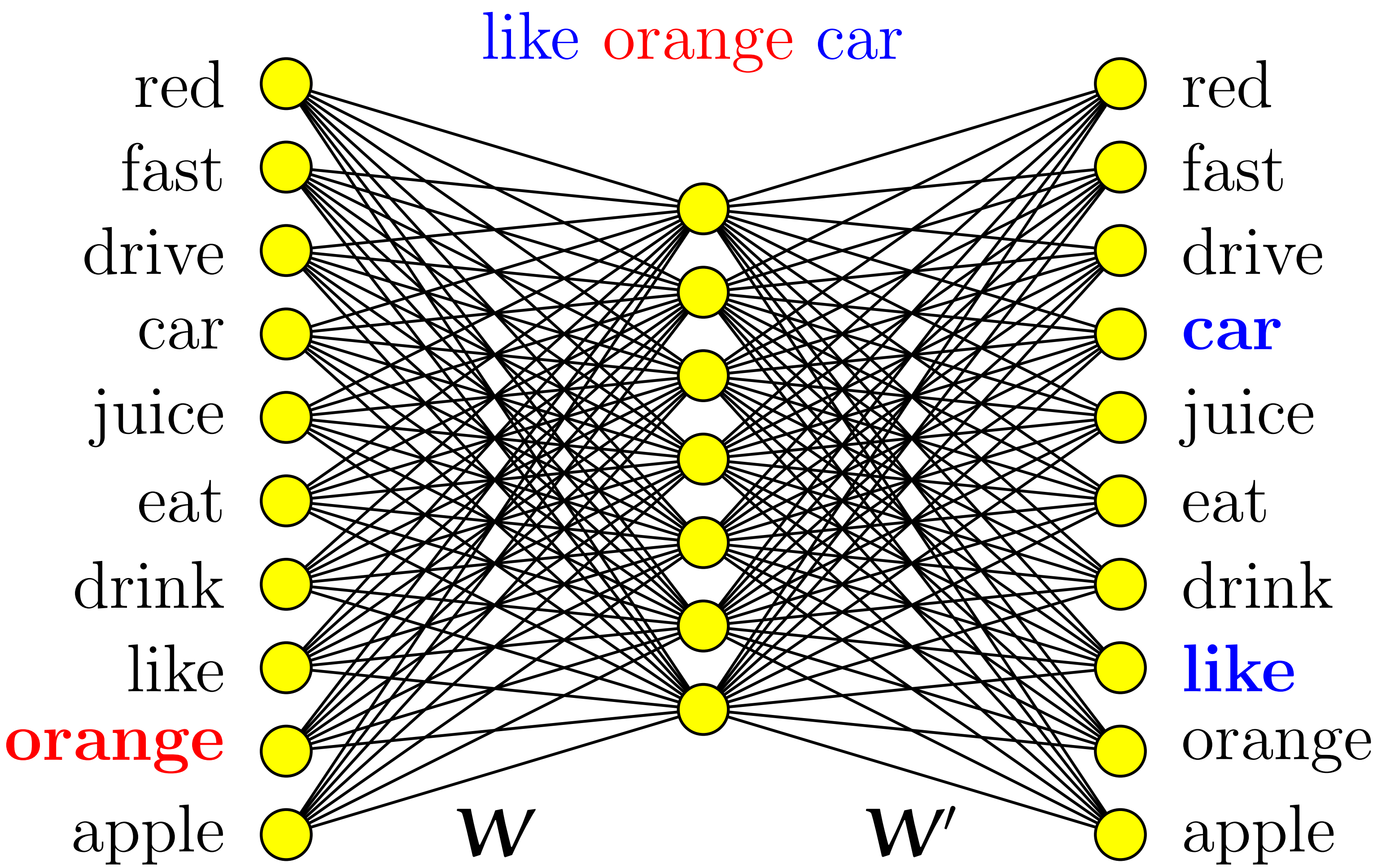

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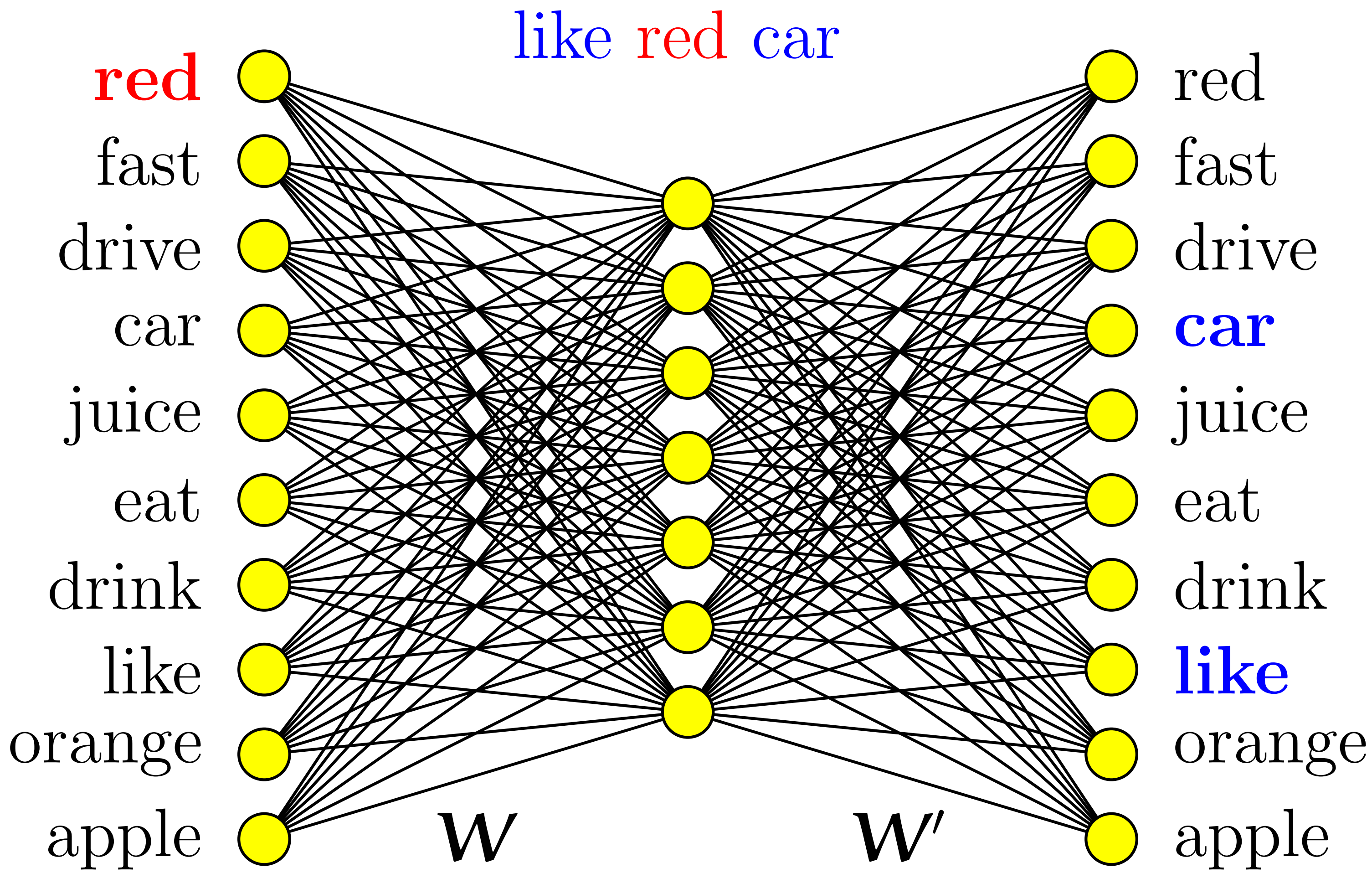

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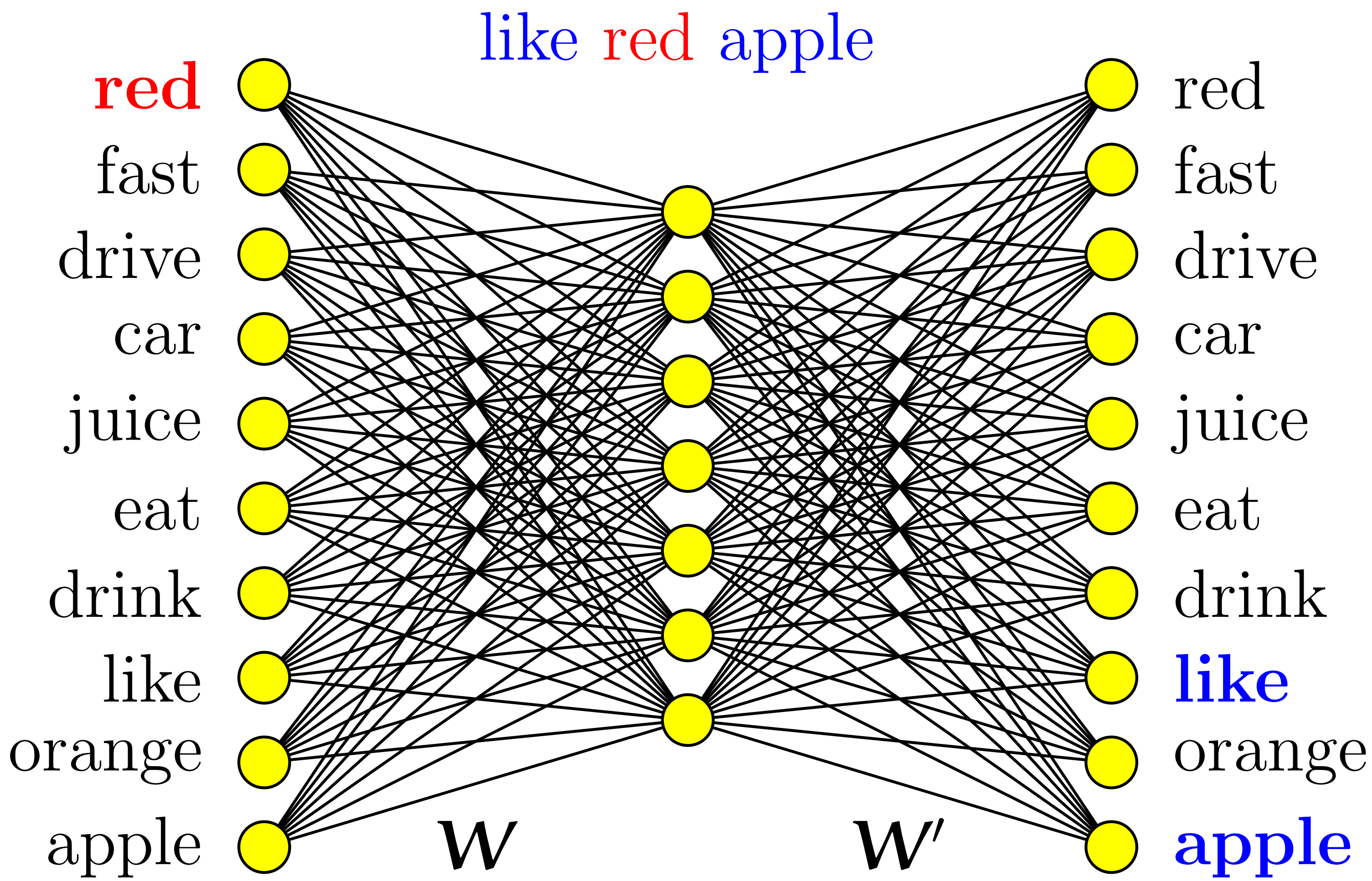

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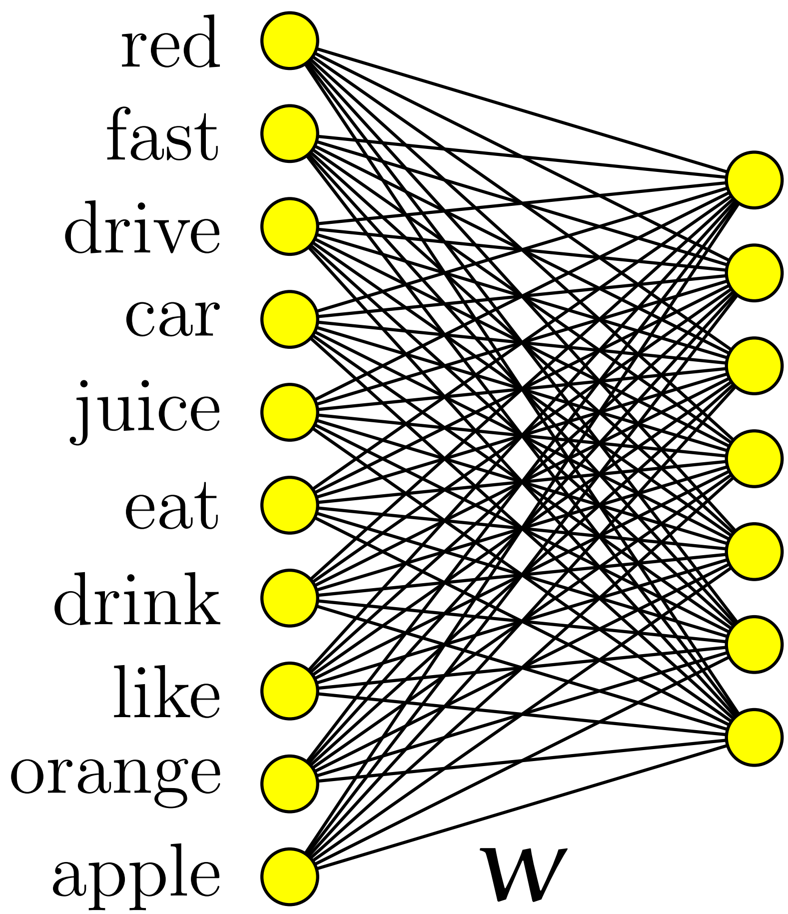
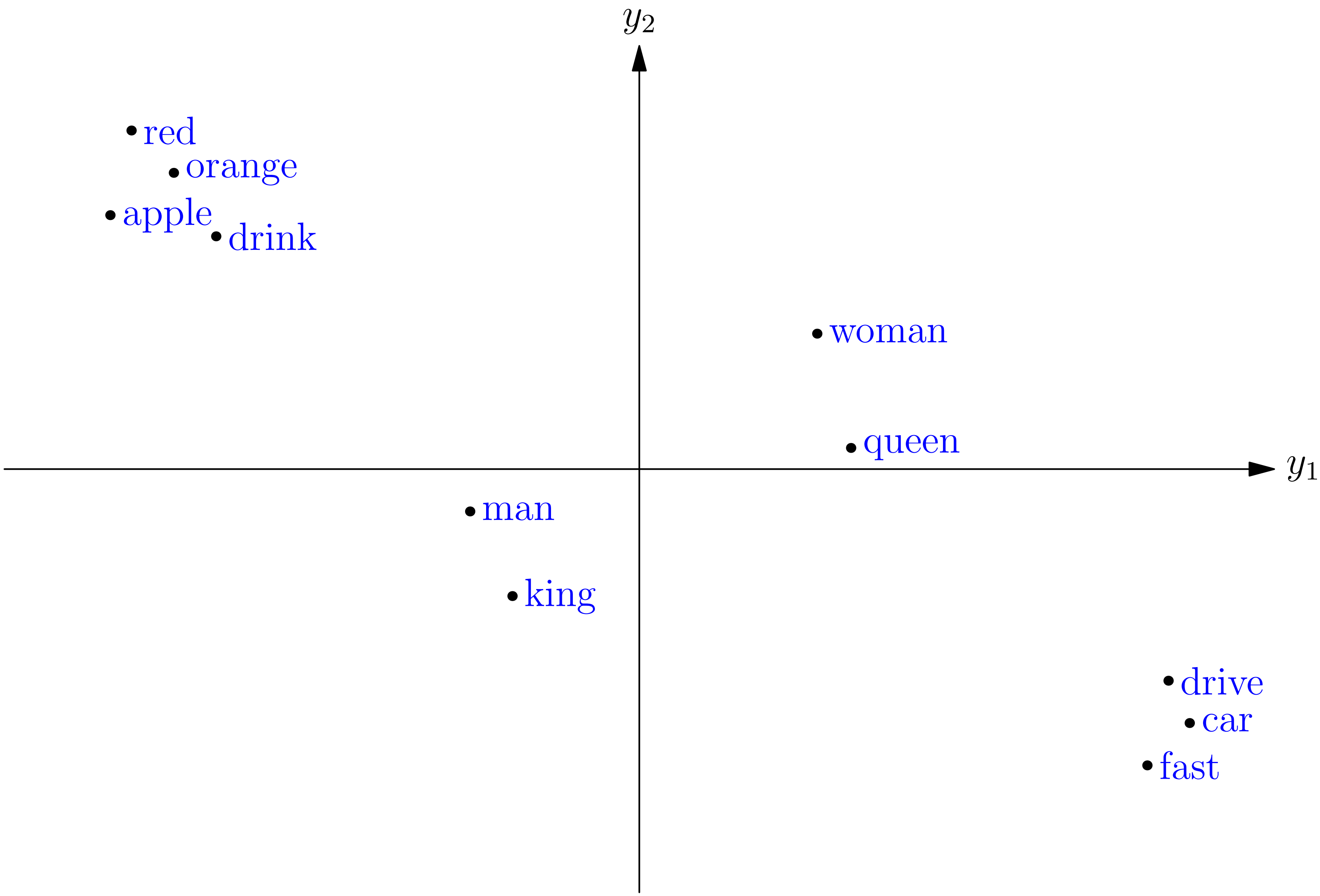


Word Embedding Space



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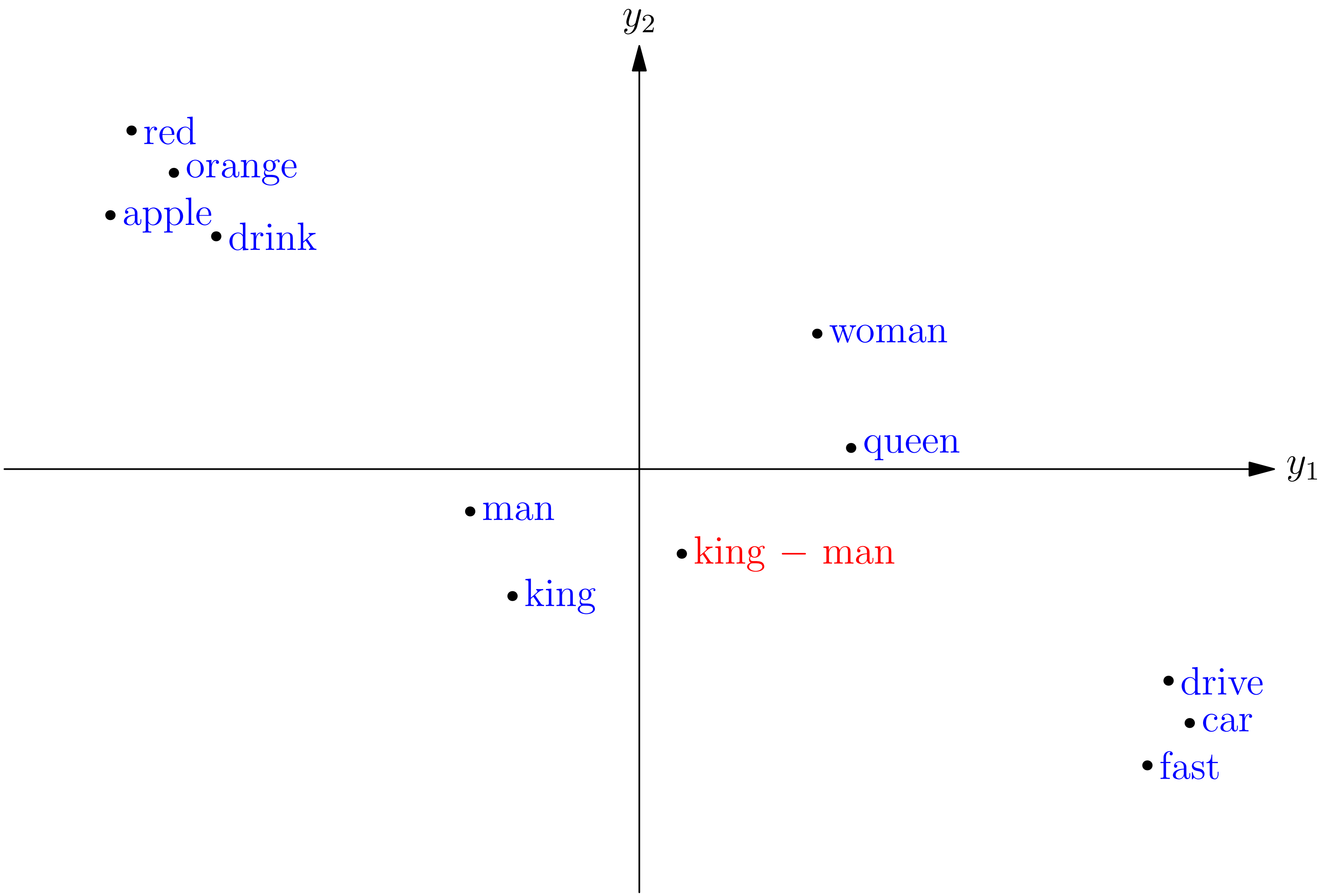

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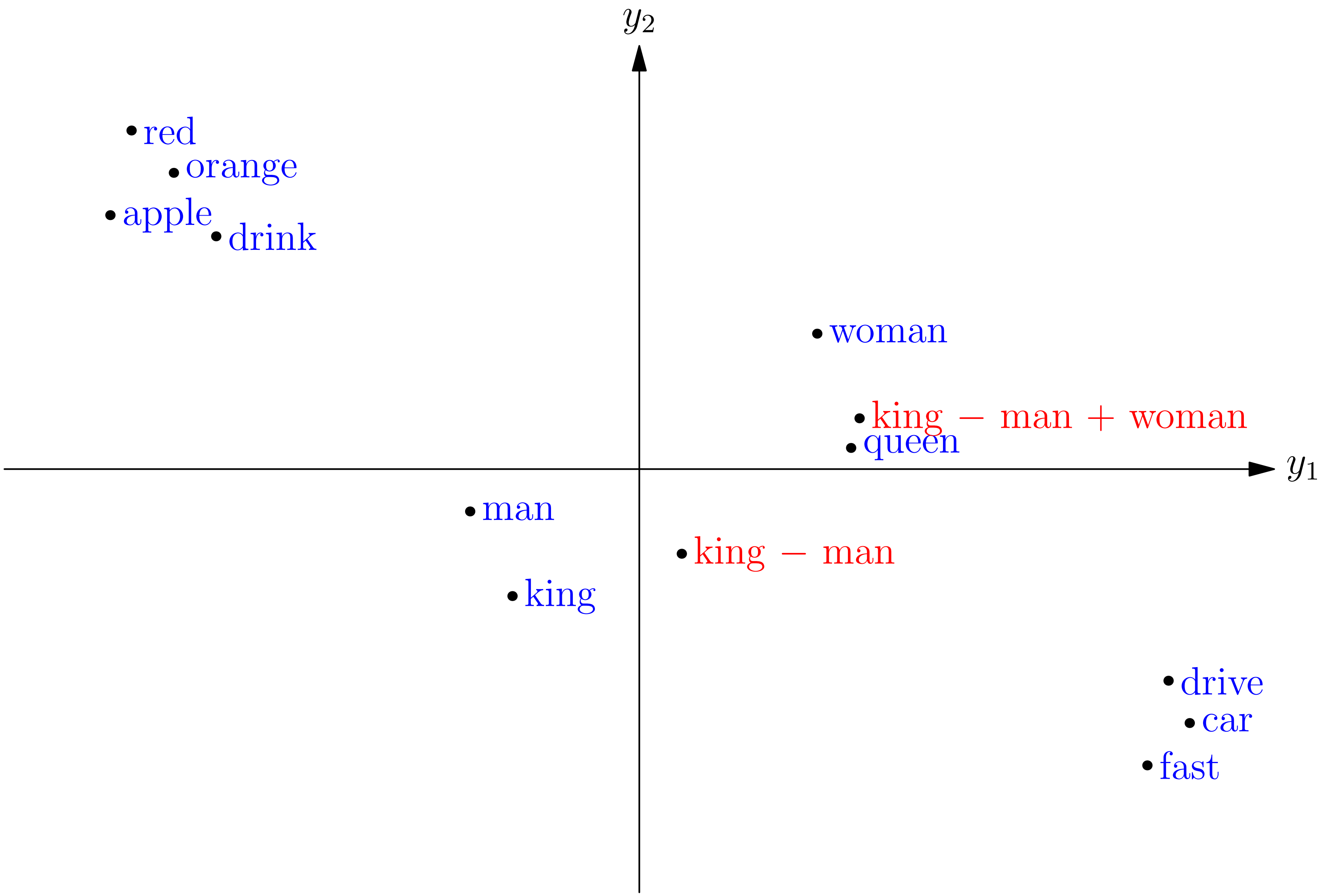
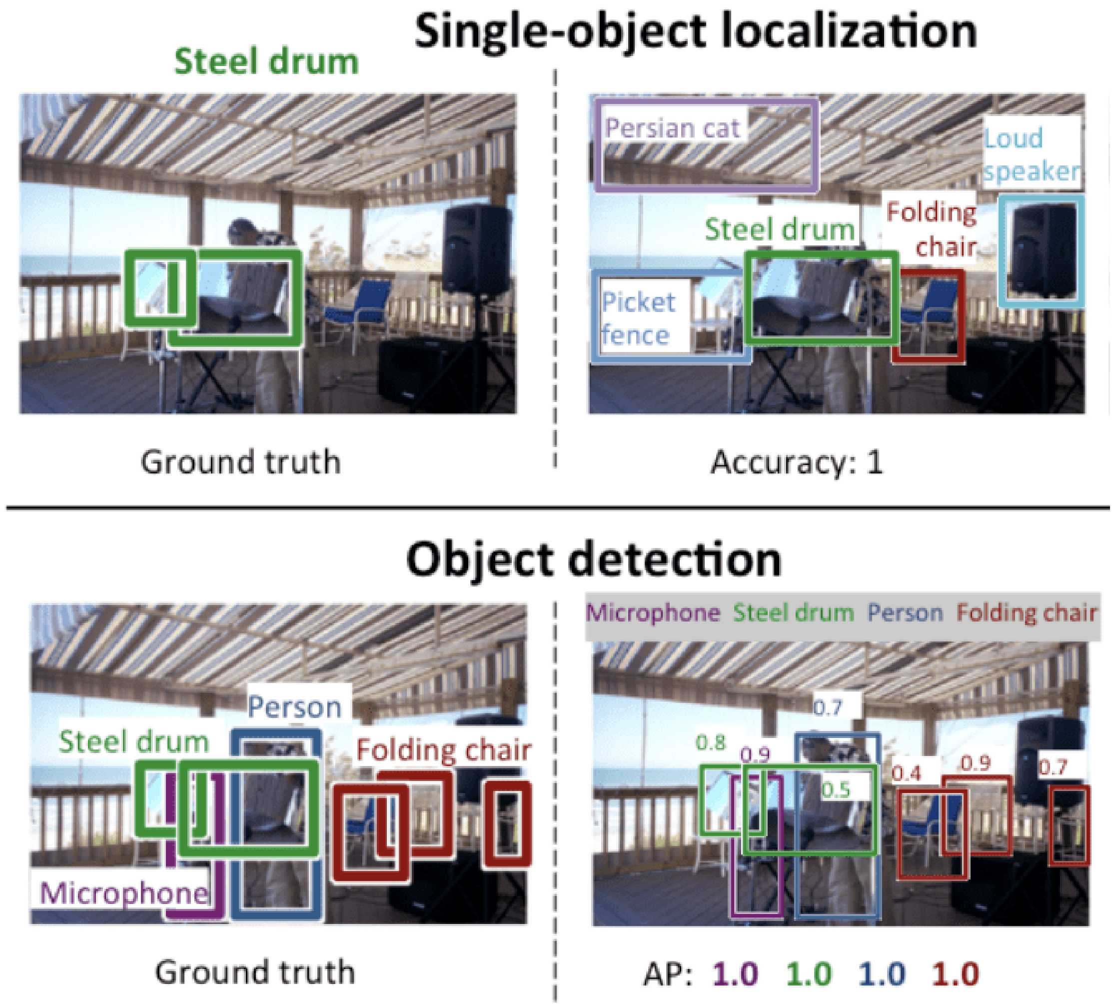


New Applications: Object Localisation



Fast-RCNN, Faster-RCNN, YOLO, Mask-RCNN

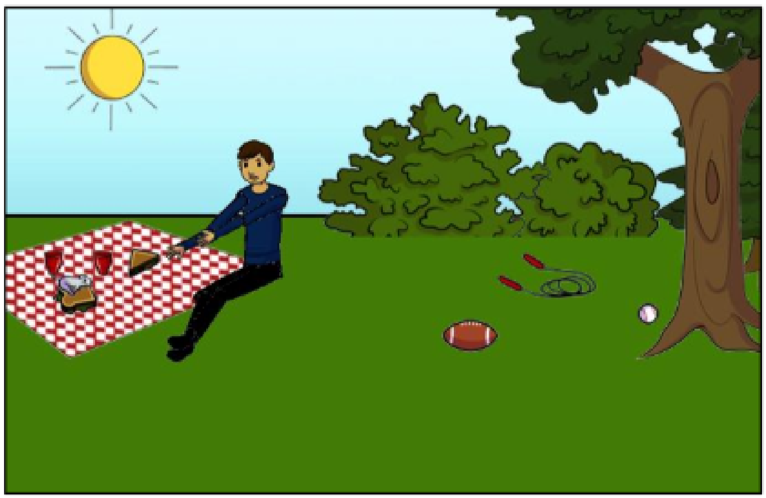
New Applications: Image Segmentation

U-Nets

New Applications: VQA

What color are her eyes?
What is the mustache made of?

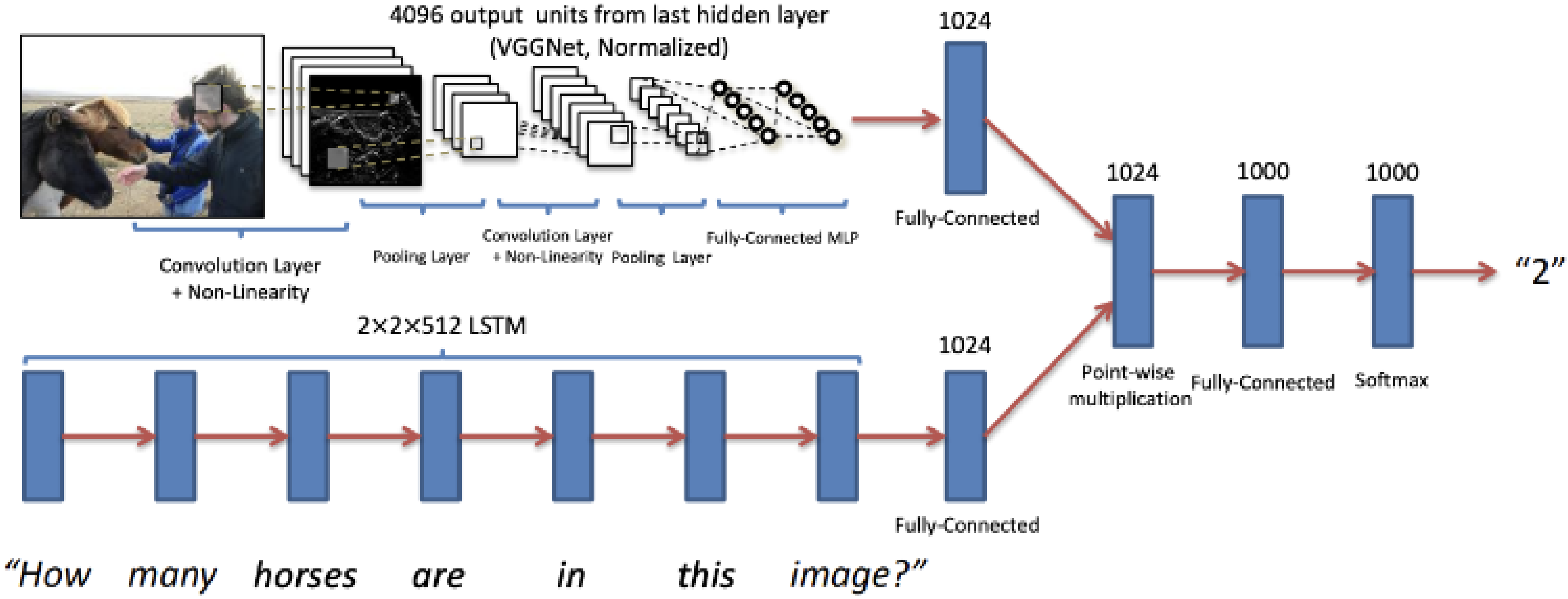
How many slices of pizza are there?
Is this a vegetarian pizza?



Is this person expecting company?
What is just under the tree?

Does it appear to be rainy?
Does this person have 20/20 vision?

Combining Networks



Towards Generative AI

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- To avoid this the community turned to **semi-supervised** learning
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Towards Generative AI

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Reinforcement Learning

- Reinforcement learning has an ancient history

- However, RL also had a reputation of rarely working
- You need to find a detailed enough representation of the state space that is small enough to exhaustive explore
- DeepMind changed that with various Go and Chess players that trained by self-play
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Multi-modality

- From early days there were attempts to couple vision with text
- Real progress came with CLIP that used a simple model that learned the correlations between parts of images and captions
- This idea has a long history
- But implemented by Open-AI, they did this on 400 million image-caption pairs
- CLIP remains a foundation for a huge number of applications from scene understanding to explainable-AI

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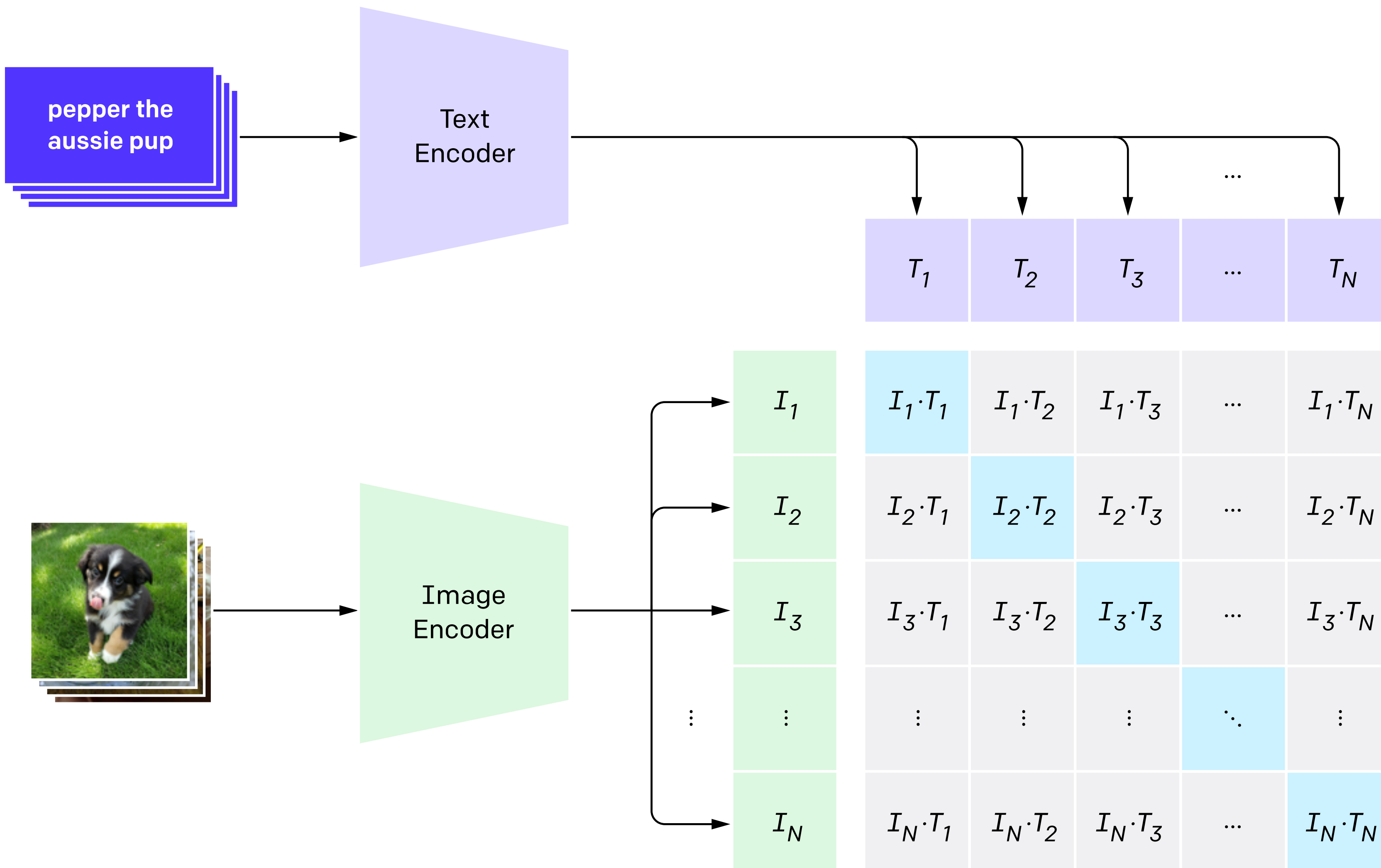
CLIP

1. Contrastive pre-training



Natural Language Processing

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- Neural networks changed the field in the 1980s
- Then Bayesian inference and statistical models took over the field
- With the rise of Deep Learning, first CNNs and then LSTMs started to compete with statistical learning models
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And Then Came Transformers

- The transformer architecture scales to large networks, but trains really well
- Also NLP turns out to be supremely well adapted to semi-supervised learning
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Attention

Words

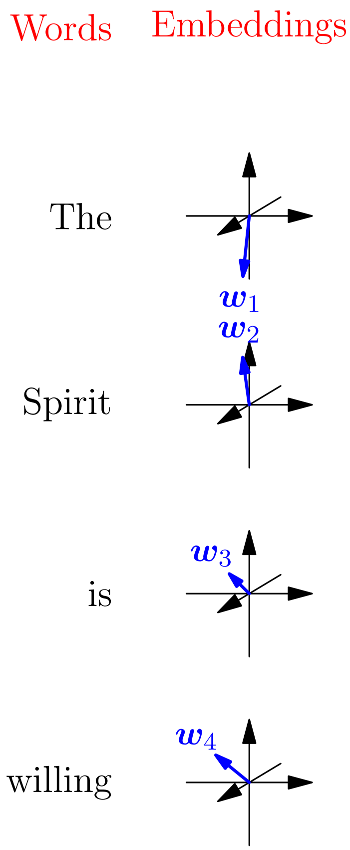
The

Spirit

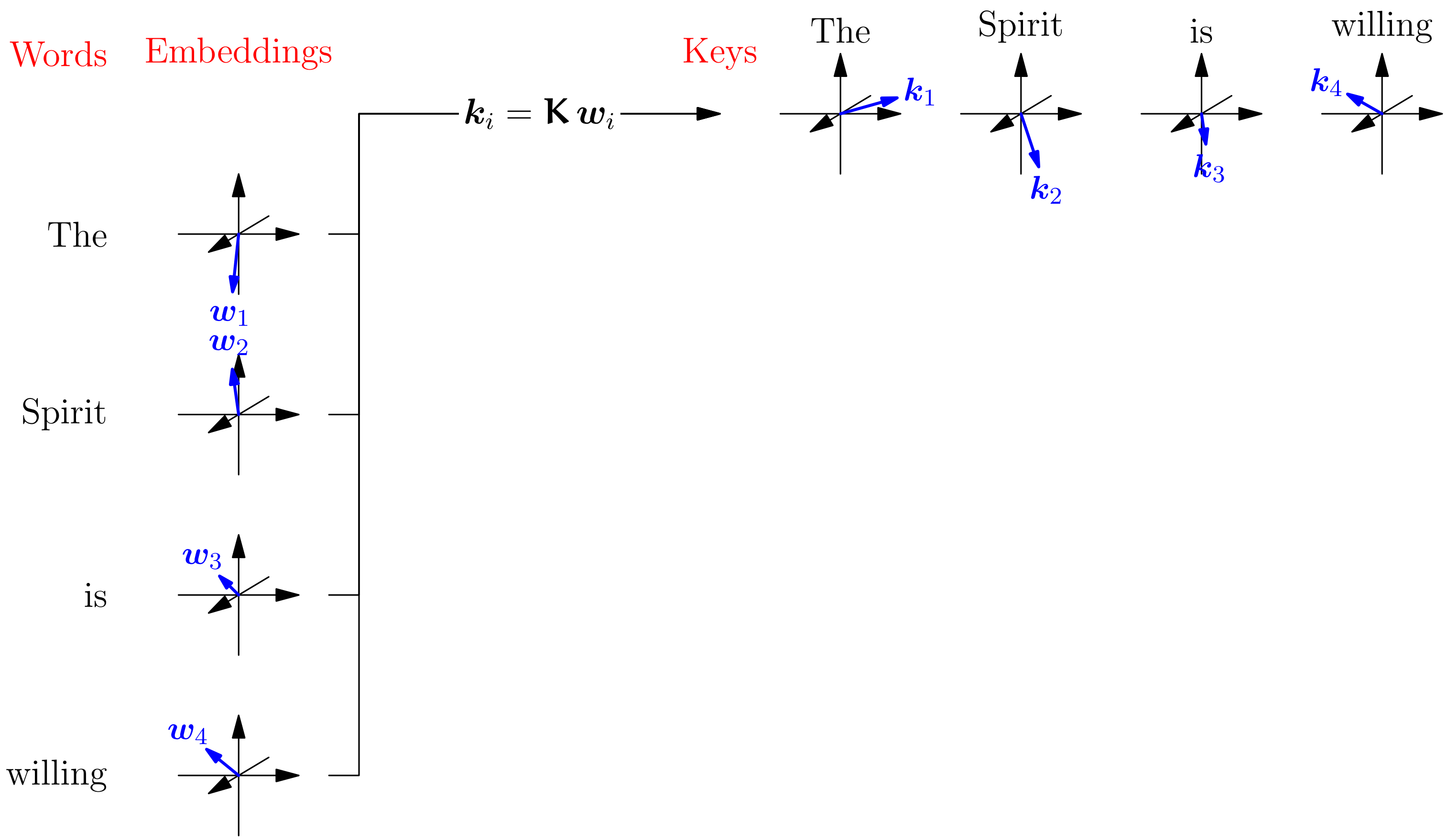
is

willing

Attention


Words
Embeddings
The
$\boldsymbol{w}_1$
$\boldsymbol{w}_2$
Spirit
$\boldsymbol{w}_3$
is
$\boldsymbol{w}_4$
willing

Attention



Words
Embeddings
The
Spirit
is
willing
$\boldsymbol{w}_1$
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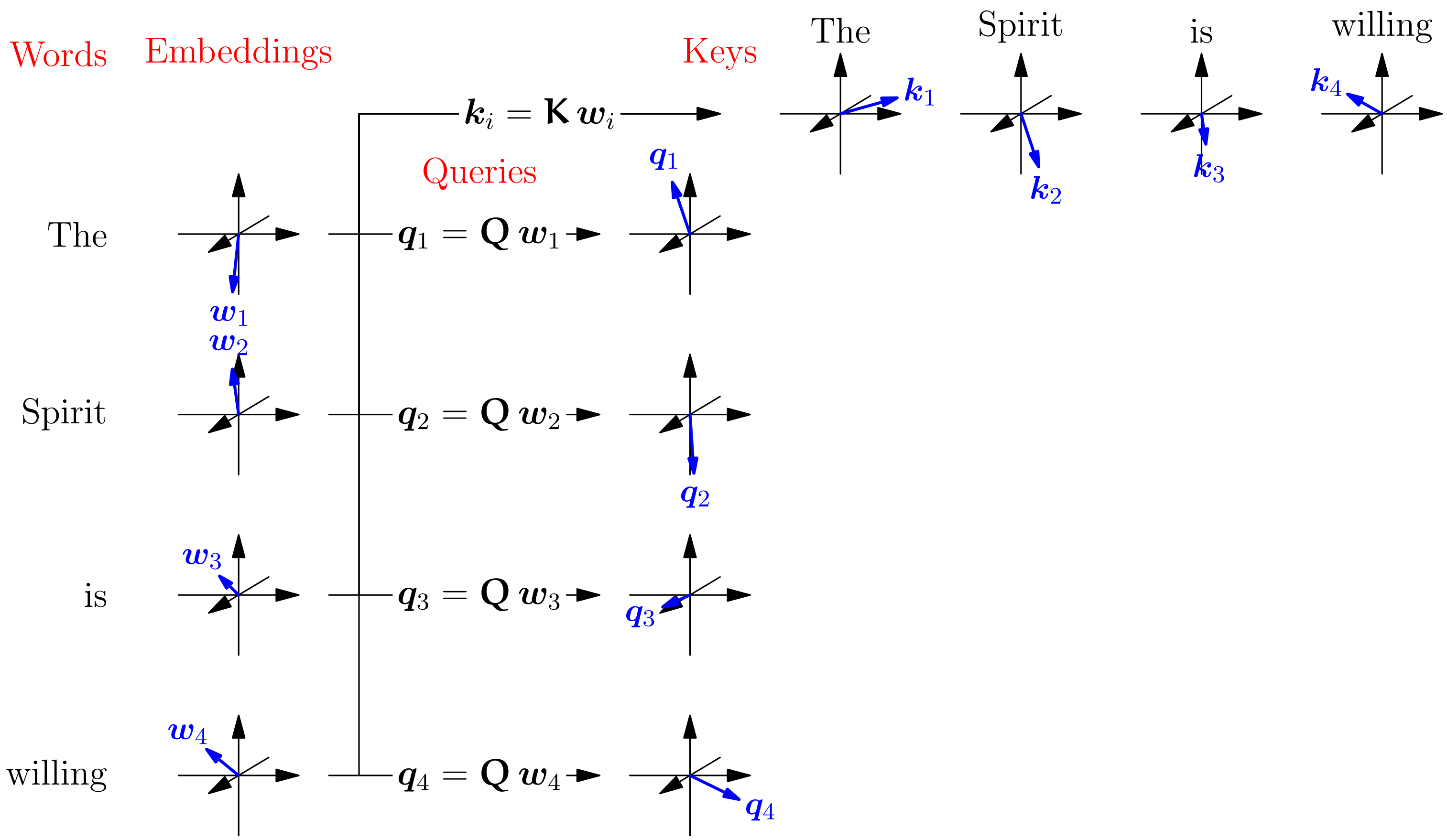
$\boldsymbol{w}_3$
$\boldsymbol{w}_4$
$\boldsymbol{k}_i = \mathsf{K}\,\boldsymbol{w}_i$
Keys
The
Spirit
is
willing
$\boldsymbol{k}_1$
$\boldsymbol{k}_2$
$\boldsymbol{k}_3$
$\boldsymbol{k}_4$

Attention


Words
Embeddings
Keys
Queries
The
Spirit
is
willing
$\boldsymbol{k}_i = \mathsf{K}\,\boldsymbol{w}_i$
$\boldsymbol{q}_1 = \mathbf{Q}\,\boldsymbol{w}_1$
$\boldsymbol{q}_2 = \mathbf{Q}\,\boldsymbol{w}_2$
$\boldsymbol{q}_3 = \mathbf{Q}\,\boldsymbol{w}_3$
$\boldsymbol{q}_4 = \mathbf{Q}\,\boldsymbol{w}_4$
$\boldsymbol{w}_1$
$\boldsymbol{w}_2$
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$\boldsymbol{w}_4$
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$\boldsymbol{k}_1$
$\boldsymbol{k}_2$
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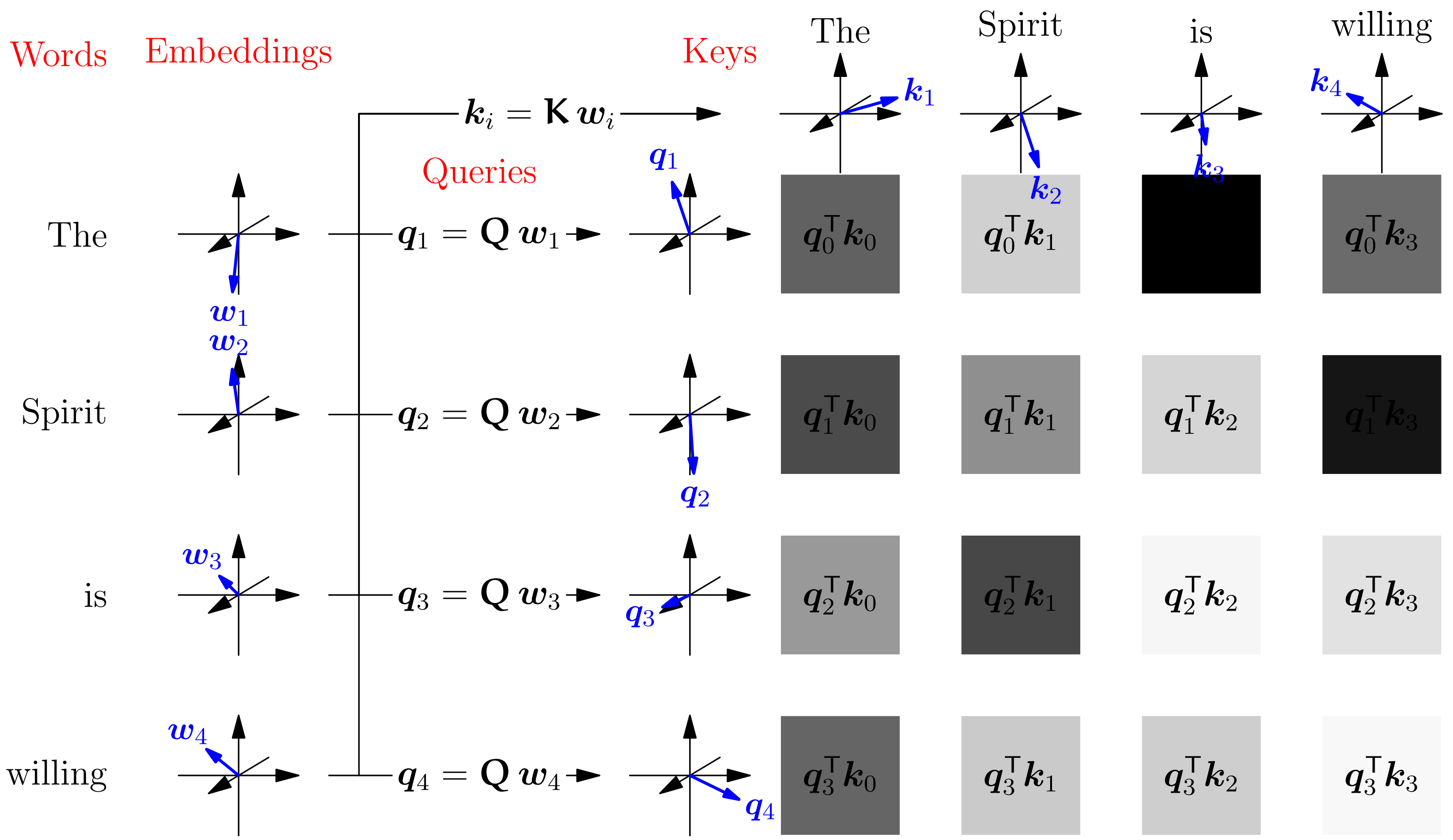
Attention



Words
Embeddings
Keys
Queries
$k_i = K\, w_i$
The
Spirit
is
willing
w_1
w_2
w_3
w_4
$q_1 = Q\, w_1$
$q_2 = Q\, w_2$
$q_3 = Q\, w_3$
$q_4 = Q\, w_4$
q_1
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q_3
q_4
k_1
k_2
k_3
k_4
$q_0^\top k_0$
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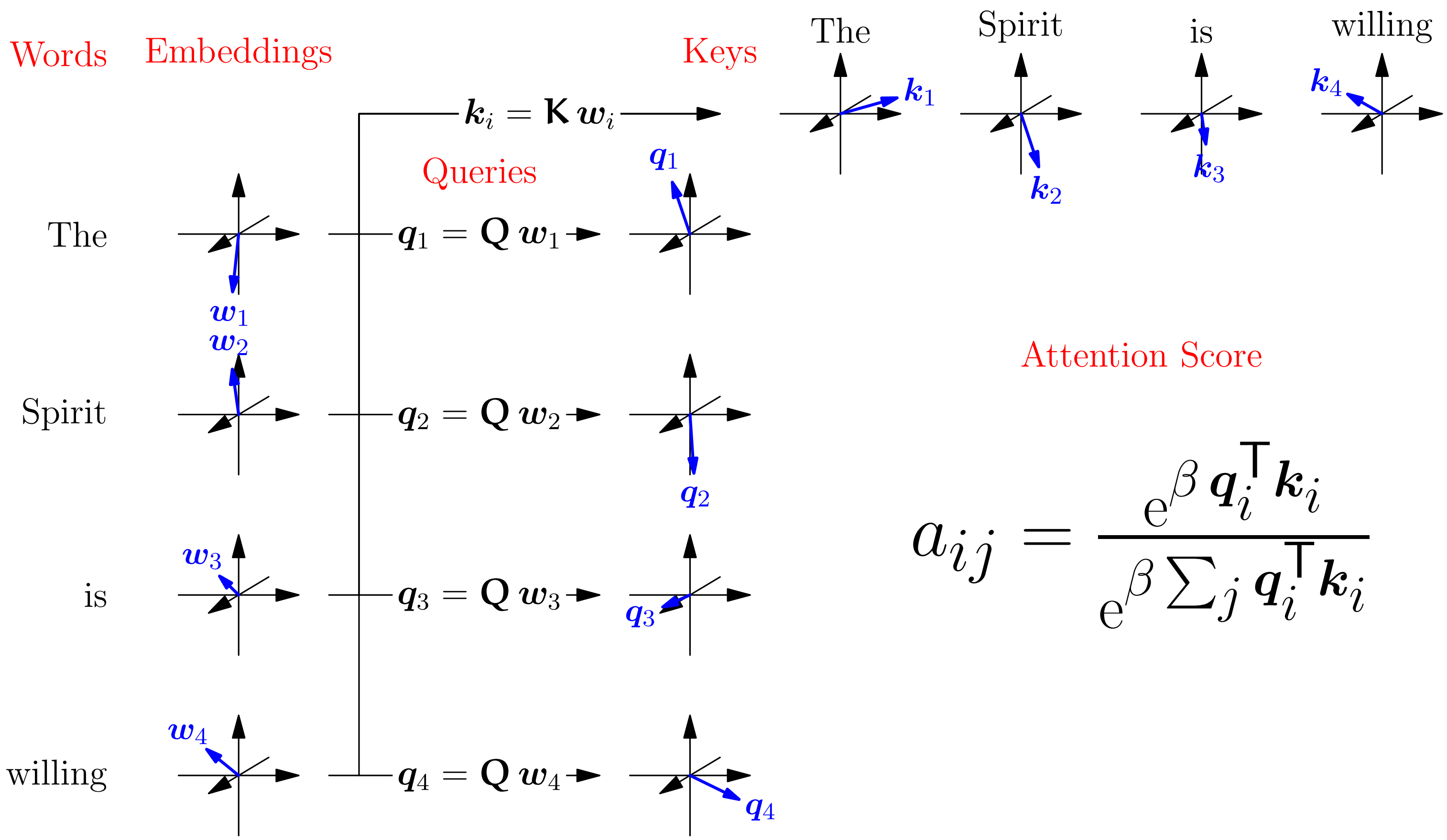
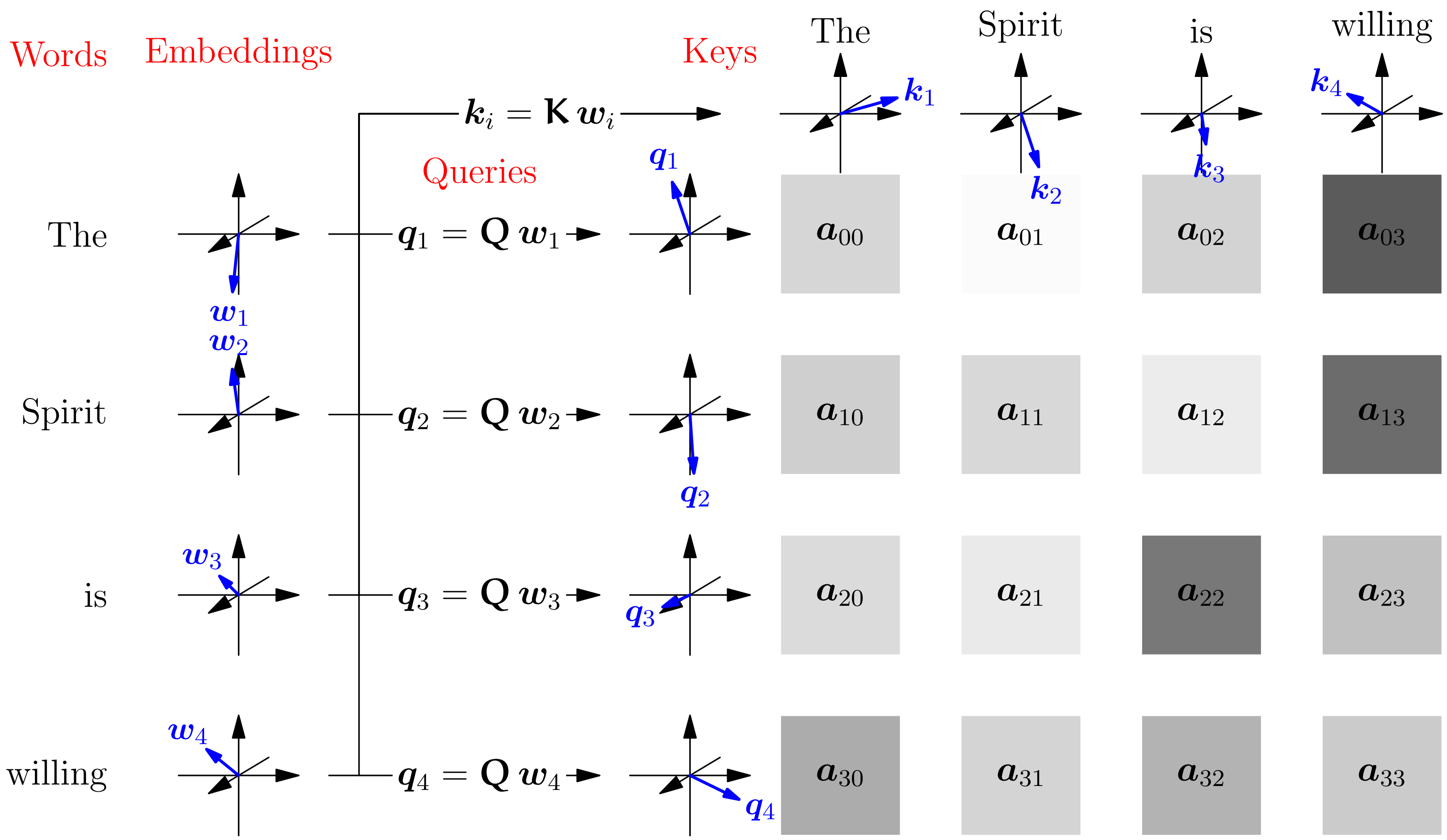
Attention



Attention Score

$$a_{ij} = \frac{e^{\beta\, \boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{k}_i}}{e^{\beta \sum_j \boldsymbol{q}_i^{\mathsf{T}} \boldsymbol{k}_i}}$$

Attention



Words
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The
Spirit
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willing
$\boldsymbol{k}_i = \mathsf{K}\,\boldsymbol{w}_i$
$\boldsymbol{q}_1 = \mathsf{Q}\,\boldsymbol{w}_1$
$\boldsymbol{q}_2 = \mathsf{Q}\,\boldsymbol{w}_2$
$\boldsymbol{q}_3 = \mathsf{Q}\,\boldsymbol{w}_3$
$\boldsymbol{q}_4 = \mathsf{Q}\,\boldsymbol{w}_4$
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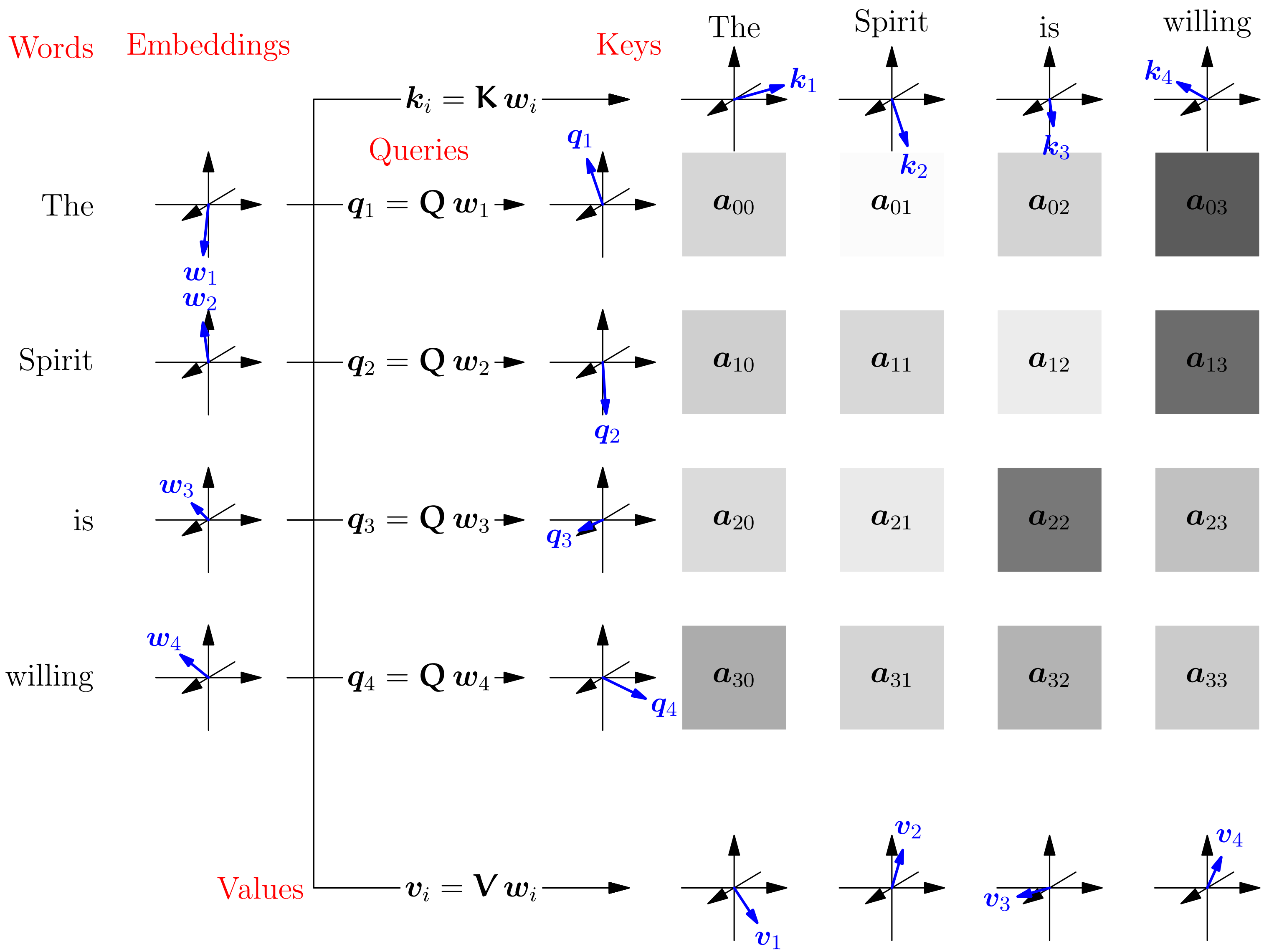
$\boldsymbol{a}_{30}$ $\boldsymbol{a}_{31}$ $\boldsymbol{a}_{32}$ $\boldsymbol{a}_{33}$

Attention



Words
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Queries
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Values
$\boldsymbol{v}_i = \mathsf{V}\,\boldsymbol{w}_i$
$\boldsymbol{v}_1$
$\boldsymbol{v}_2$
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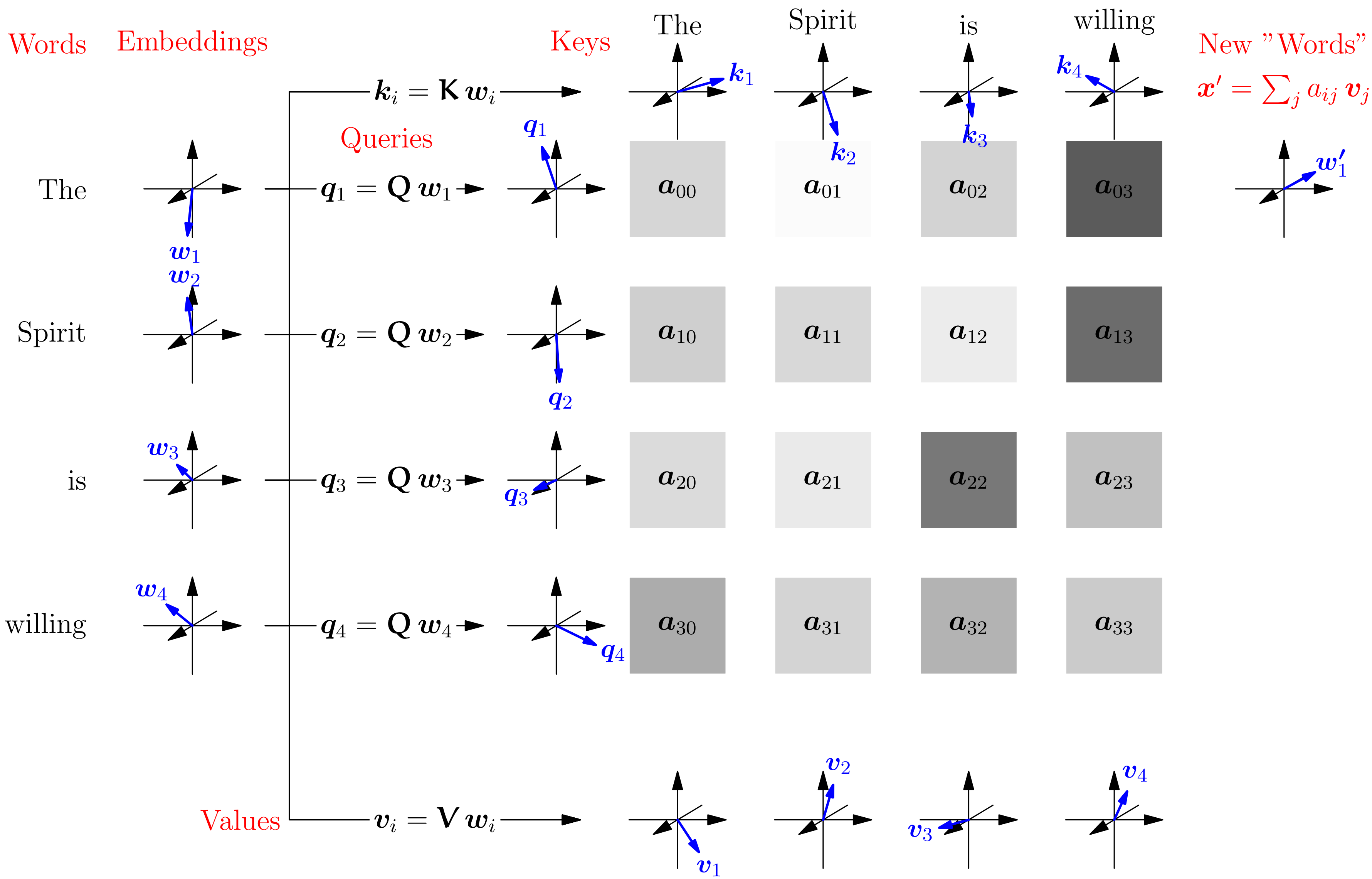
Attention



Words
Embeddings
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The
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New "Words"
$\boldsymbol{x}' = \sum_j a_{ij}\, \boldsymbol{v}_j$
$\boldsymbol{k}_i = \mathsf{K}\, \boldsymbol{w}_i$
$\boldsymbol{k}_1$
$\boldsymbol{k}_2$
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Queries
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$\boldsymbol{q}_1 = \mathsf{Q}\, \boldsymbol{w}_1$
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$\boldsymbol{w}_1'$
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$\boldsymbol{w}_2$
Spirit
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$\boldsymbol{w}_4$
willing
$\boldsymbol{q}_4 = \mathsf{Q}\, \boldsymbol{w}_4$
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$\boldsymbol{a}_{30}$
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$\boldsymbol{a}_{33}$
$\boldsymbol{v}_2$
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Values
$\boldsymbol{v}_i = \mathsf{V}\, \boldsymbol{w}_i$
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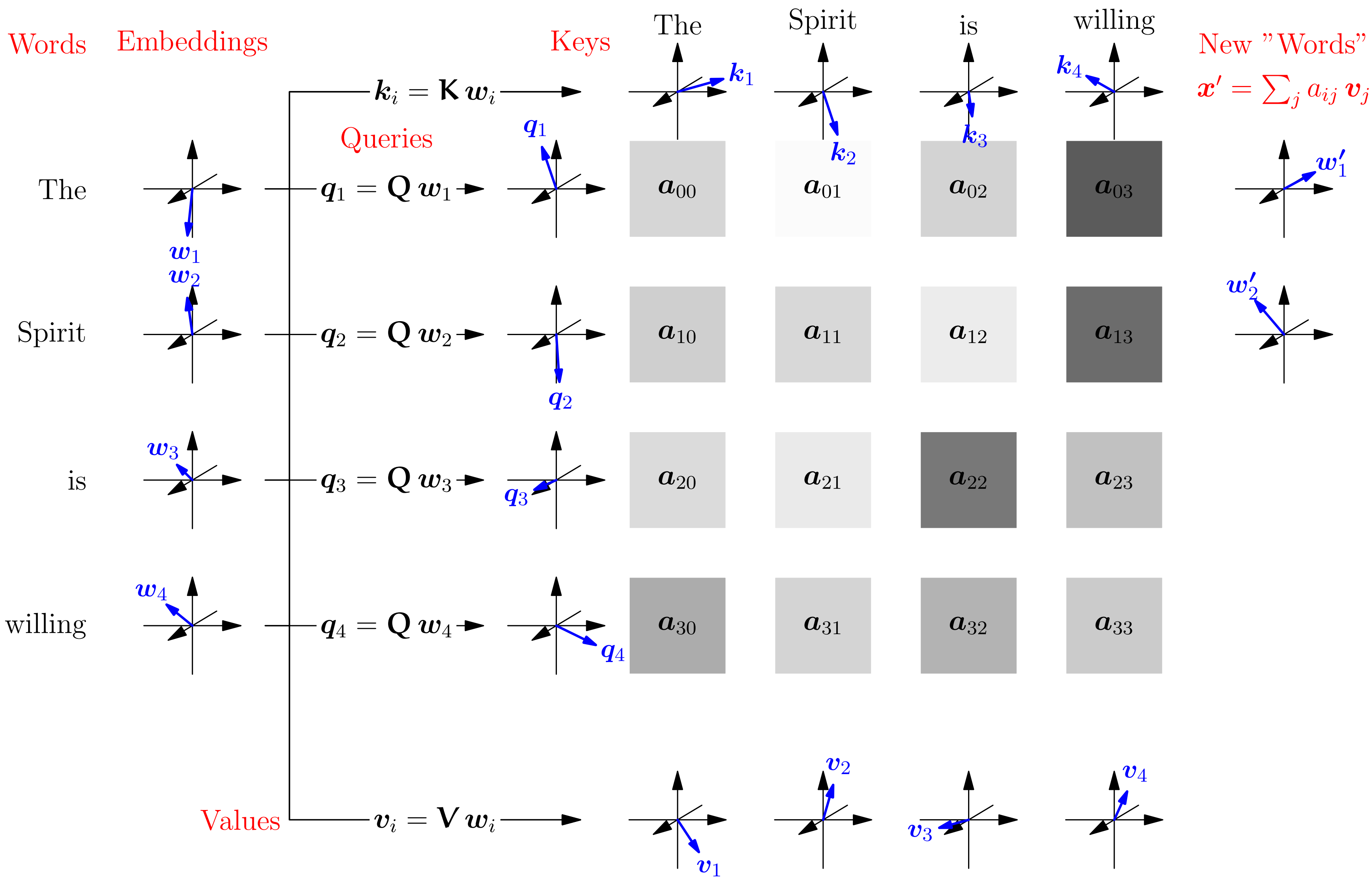
$\boldsymbol{v}_1$

Attention



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$\boldsymbol{w}'_1$ $\boldsymbol{w}'_2$

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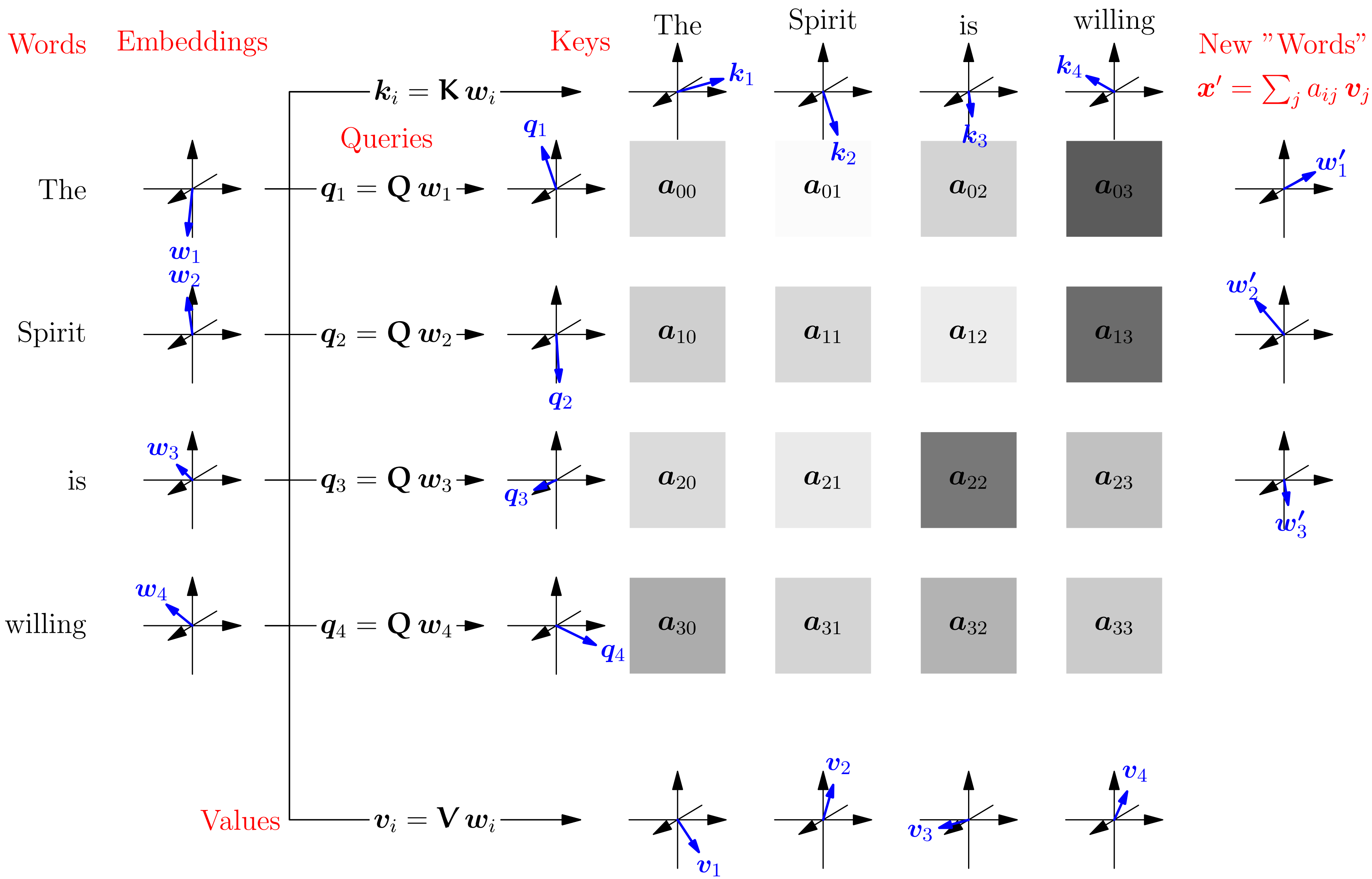

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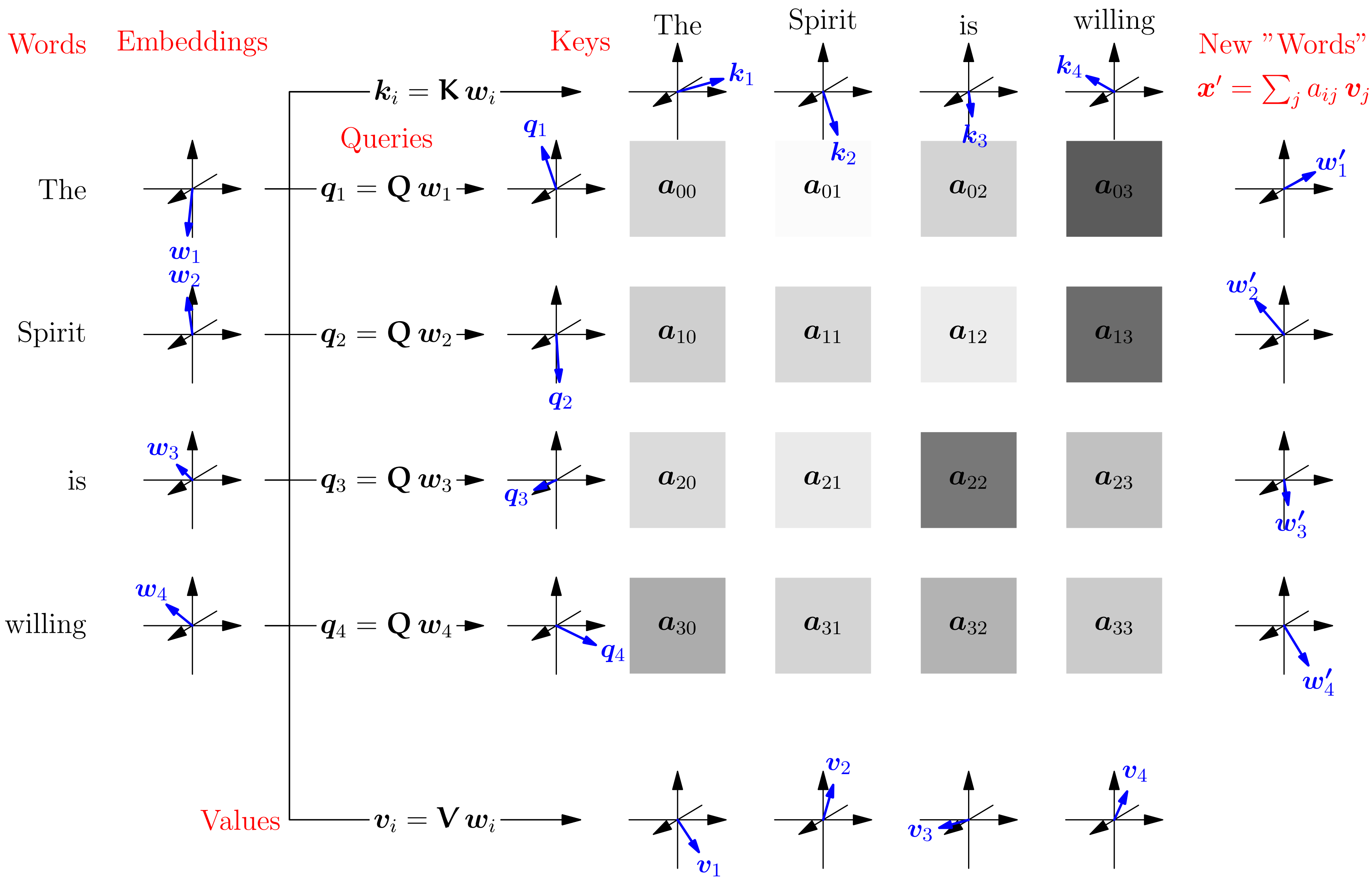

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$\boldsymbol{w}_1$ $\boldsymbol{w}_2$ $\boldsymbol{w}_3$ $\boldsymbol{w}_4$
$\boldsymbol{a}_{00}$ $\boldsymbol{a}_{01}$ $\boldsymbol{a}_{02}$ $\boldsymbol{a}_{03}$
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Values
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Training is More Important than Architecture

- The transformer model has its limitation
- Issues include, embeddings and context window size
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- Using huge, well curated datasets, chain-of-thought reasoning, reinforcement learning, etc. makes the real difference

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- That is, if you translate the input the output is translated
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- But they have the advantage of understanding combinations of features
- Visual Transformers for a short while seemed to be replacing CNNs
- Today we try to gain the advantages of both architectures

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Ethics

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- Now ethics in AI is huge
- Issues around bias, replacing peoples jobs, accountability are all now prescient
- Explainable AI has emerged as an important field, albeit one with very imperfect solutions

AI and Big Business

- Because huge models trained on huge datasets now rule the roost, large companies play a disproportionate role in driving AI
- They tend to only work on problems where there is money
- Although, some companies, such as DeepMind, get involve with non-profit prestige projects
- There remain many problems requiring cheap development

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